



A new image encryption based on bit replacing, chaos and DNA coding techniques

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Abstract

Millions of confidential images are transmitted every day by people through the multimedia systems of the internet. Such confidential information in the images may belong to the political, business, medical or military authorities. Protecting the privacy and integrity of these images in the era of the internet and multimedia technologies has drawn more attention by the research community. Image encryption is one of the most important tools to secure digital images from the unauthorized access and malicious manipulation. In this paper, a new digital image encryption method is proposed based on bit replacing technique, chaotic systems and DNA coding algorithm. It aims to protect the confidentiality and privacy of the digital images sent over unsecure open channels. Firstly, in this method each pixel of the image is converted to its corresponding binary sequence comprising of zeros and ones bits. Then, the zero bit is replaced by (1 and 0) bits and the one bit is replaced by (0 and 1) bits. Two different images are eventually generated by repeating consistently the replacing operation for all bits of the image pixels. Secondly, the generated images are encrypted using high dimensional chaotic systems based on the principle of permutation and diffusion processes in objective to vary the positions and values of the digital image pixels. Thirdly, the resultant encrypted images are encoded by adopting DNA algorithm rules and then these images are merged by exploiting DNA addition operation. Finally, the coded DNA images are decoded to obtain the output encrypted image. The numerical and visual simulation results confirm that the proposed method is sufficiently robust and secure against several known attacks in comparison with the state-of-art approaches. Also, the conducted experiments show significant improvement in terms of entropy, correlation, differential, discrepancy metrics, key space and computational speed analysis performance parameters. To sum up, the proposed approach

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produced large secret key space of (2^{747}), a comparable differential analysis performance NPCR (99.61%) and UACI (34.61%) and passed all security and randomness tests.

Keywords Digital Image Encryption, Bit Replacing Technique, Chaotic Systems, DNA Coding

1 Introduction

Securing digital images is of the utmost importance nowadays with the fast advances in the multimedia and internet technologies. One of the most proved successful methods for safe transmission of images through the internet is the cryptography. It is used to transform the plain image into the scrambled image by confusion and diffusion operations in the encryption process at the transmitter, while the decryption process is used to recovering the original image at the receiver. The cryptographic key is used by encryption and decryption processes to secure digital assets and communications [1, 4, 24, 61]. Conventional cryptography is categorized into two major categories based on the type of cryptographic keys used. Symmetric algorithms are the first class of cryptographic algorithms that use same shared key between two communicating parties, while asymmetric algorithms (or public key algorithms) is the second class of cryptographic algorithms that use different two pairs of keys [62]. However, symmetric algorithms like Data Encryption Standard (DES) and Advanced Encryption Standard (AES), and asymmetric algorithms like RSA are more adequate for text security than digital images security due to the substantial characteristics of images such as strong relationships among neighboring pixels, large redundancy and bulk information capacity. Hence, it is substantial to develop new cryptosystems, mechanisms and methods to protect the confidential data of image to cope with the fast advances in the communication systems [2, 3], since the traditional encryption algorithms are inadequate for real time image security in the telecommunications systems [21] and.

Chaos based cryptographic algorithms proved their impressive performance in securing digital images. Such performance is attributed to the distinctive characteristics of chaotic systems including nonlinearity, high sensibility to initial parameters, pseudo randomness and non-periodicity. Scrambling and diffusion are the main processes in any encryption algorithm. The scrambling process is used to alter the pixel's locations, while diffusion process is used to vary the pixel's values. Usage either of these processes (i.e. scrambling or diffusion) separately in image cryptosystems will weaken the security, hence both processes should be used together in the image cryptosystem to satisfy required security requirements [6, 23]. Utilizing low dimensional chaotic systems (i.e. 1D or 2D) is the major threat to these processes that could lead to the breaking of the cryptosystem because high possibility of guessing cryptographic key, short periodicity and little key space. However, the chaotic systems of higher dimensions (3D or more) have multiple parameters, complicated structure, large key space and high sensibility that making breaking image cryptosystem is difficult task [16, 17, 44].

DNA computing is a modern field of the cryptography which possesses several interesting properties like ultra-low power consuming, enormous storage and huge parallelism. It is the new image cryptosystems which has four complementary rules including addition, subtraction, XOR and XNOR DNA processes. DNA encryption represents modern biological technique within DNA computing in which DNA is used as a data carrier. The DNA encryption method can be partitioned into two main steps. In the first step, converting the input image into the DNA sequence matrices and producing the key stream using DNA encryption rules, whereas in the second step, the input image is encrypted using DNA sequence processes and the

generated key streams. Also, the encoded DNA sequence matrices are transformed into encrypted image using DNA decryption rules [13, 55, 56].

There are many approaches proposed for the digital image security. For example, the researchers in [55] utilized the rules of DNA coding and chaotic maps to encrypt color images. First, the original color image is divided into three color components R, G and B, then these three color matrices are encoded by employing DNA rules. In another research [17], the structure of permutation/diffusion in the chaotic system is used to encrypt digital images. The authors in [43] utilized cycling chaotic map and DNA technology for pixel positions permutation where masked DNA is employed for pixel values alteration. In the same direction, the authors in [56] developed a color image security algorithm using DNA operations and chaotic systems. The encoded image in the earlier scheme is obtained by applying DNA rules on the plain image to get the scrambled image. In [52], random integers are produced by cycle shift technique to shuffle the plain image. Next, the encrypted image is acquired via the secret keys generated from the applied chaotic system. Furthermore, the researchers in [11] adopted the chaos theory to establish the relationship between the encryption algorithm and the plain image in aim to enhance the performance in terms of selected keystream sequence group mechanisms.

Logistic, Sine and Chebyshev maps are the three one dimensional chaotic maps which are used to construct the robust image security approach by authors in [41]. However, 5D chaotic system is adopted to secure grayscale images in another research by doing permutation and diffusion processes at the bit-level/pixel-level [31]. Another study in the same direction is presented by authors in [23] to improving the connection between pixel locations shuffling and their values and also the study of authors in [44] is introduced to secure color images by 3D chaotic system which it obtained by integrating logistic and piecewise maps. Another an image encryption system is proposed by authors in [32] by utilizing hash function and hyper Lorenz system and the secret key is represented based on the initial values of the chaotic system. Encryption process in [42] is accomplished by two stages, confusion and scrambling operations. These two operations are executed in this algorithm by implementing XOR function between the original image pixels and random numbers generated from applying double Logistic map. In addition, the key stream in [19] is produced by employing Logistic and Lu chaotic systems for mixing and scrambling the information of the original image. Swapping technique is used in the permutation phase to scramble the pixel's locations, whilst key stream production mechanism is utilized in the substitution phase to modify the pixel's values.

A new 2D chaotic map is proposed in [57] by integrating both Sine and Henon maps. The output encrypted image in this method is obtained via two operations, first one is image permutation by applying chaotic map and the second is the image diffusion by applying DNA technology. The researchers in [65] used symmetric cryptography combined with chaotic systems and 3D permutation mechanism to protect the image data during transmission. The plaintext image in this system is permuted at the sender by using 3D permutation technique, and then XOR operation is performed on the resultant permuted image to get the final encrypted image. Also, a new chaotic system is developed in [51] depends on staged Logistic and Tent maps combined with composite map to encrypt color images. These maps used to generate the initial keys needed for Arnold map for image pixels scrambling stage, whilst Hopfield neural network is used for image pixels' diffusion stage. By applying XOR operation between the key and the scrambled image in the final stage, the encrypted image is obtained. 3D chaotic map is suggested in [50] for encrypting color images. The corresponding R, G and B channels of the original color image are XORed with the random numbers produced from the suggested chaotic map. The process is started by separating the image into 4×4 and 16×16 blocks, and then the keys of new map are permuted with those

blocks. After finishing the permutation process for all image blocks, the encrypted image is earned. The input image is scrambled and encrypted in [27] by employing a new 1D chaotic map named coupled Sine map. The presented image cryptosystem in this work is evaluated by relying upon different security analyses.

The developed method for the image cryptosystem in [18] depends on arbitrary precision arithmetic of chaotic generator and it characterized by low cost and possessing large key space. Block permutation and stochastic insertion mechanisms are merged in [46] for encrypting gray and color images. Pseudorandom sequences are generated from applying 5D chaotic system, then these secret keys are combined with hash function for producing the initial values of this cryptosystem. Hyper chaotic map, Rossler and Piece wise linear systems are joined together and then the hash algorithm is used to confuse the pixel locations and diffuse their values [66]. Hence, high security is achieved by exploiting confusion/diffusion characteristics on the plaintext information. A new model for image encryption is introduced in [53] by utilizing a new chaotic system that generated from concatenating Arnold and Logistic maps. The suggested chaotic system in this cryptosystem is used to provide the key stream for performing both diffusion and confusion operations. A hybrid image cryptosystem is proposed in [48] based on Lu and Henon chaotic maps, spiral scanning technique and overlapping partition technology. The original image is partitioned initially to overlapping blocks, then the block pixels are shuffled by employing spiral scanning mechanism. Next, the secret keys are generated from the used chaotic maps to compute the secret matrix that is combined with the shuffled image through XOR process to get the encrypted image. Generalized mechanism of chaos game is presented in [8] for encrypting digital images. The interest area of this method is based on the fractal technology and bases number of the introduced chaos game. Five chaotic systems namely Logistic, Rössler, Chen, Tinkerbell and Henon maps are utilized in [21] on a PIC microcontroller to cipher images over wireless telecommunication systems.

The input image in [58] is converted to scrambled image through confusion operation by altering image pixels' positions. This step is followed by diffusion operation on the obtained scrambled image to get the final output image. These two steps are depending on pseudorandom sequences produced via Lorenz system. In [60], image ciphering approach based on new suggested chaotic system and DWT is proposed. The actual image is decomposed in this scheme via DWT mechanism to sub bands. Then, permutation and substitution characteristics are fulfilled by utilizing parameters and states of the presented chaotic map. 4D hyper chaotic Lorenz system and DNA cryptography are utilized in [36] for image encryption. The secret key is generated by the described chaotic system and DNA algorithm is used for image coding through a series of processes like scrambling and addition/XOR operations. Color image encryption scheme based upon calculations of DNA coding and chaotic maps over Galois field is introduced in [26]. The initial keys are produced via chaotic system and DNA coding is applied to achieve the permutation process of this method, while Galois field is employed to improve the impact of pixel diffusion. In [28], chaotic encryption is applied first on the R, G and B channels of the input color image based on selection mechanism. Next, DNA coding method is utilized on the encrypted channels to obtain the final encrypted image. DCT and selective encryption method are implemented in [22] to encode color image. By utilizing these two techniques, the computation complexity and encoding/decoding time is greatly reduced.

The medical image is enciphered in [15] by executing confusion/diffusion concept. Confusion step is attained by applying high-speed scrambling technique, while diffusion step is realized by employing pixel adaptive technology. Tent, Sine and Logistic maps are mixed in [54] to design a new chaotic system. After that, the produced chaotic model is used in this scheme for image

encryption via confusion/diffusion architecture. An image encryption method that depends on multi-dimensional chaotic maps and interest region technique is presented in [38]. Logistic, Unified and Henon chaotic maps as well as oriented gradients histogram mechanism are utilized to accomplish the diffusion process for obtaining the encrypted image. Ultimately, in [69], DNA coding rules and chaotic systems are joined to propose a grayscale image encryption method. The confusion stage on the image pixels is carried out by random sequences of Logistic-Sine system and Henon map. This process is followed by applying DNA mechanism and XOR operation of DNA is implemented on the confused image in the diffusion stage.

To remedy the existing security and efficiency limitations in the above approaches for image security, this research introduces a new secure and efficient image cryptosystem that merges both high dimensional chaotic systems and DNA sequence encoding operations. Adopting the proposed techniques in the presented method will enhance the cryptosystem robustness against attacks. Firstly, the input image in this approach is transformed to binary form such that each pixel consists of eight bits. Then, each 0 bit is replaced by one and zero bits, respectively; while each 1 bit is replaced by zero and one bits, respectively. This operation is repeated for all the pixels in the original image to produce two different images. Secondly, the resultant images from the first step are permuted by using 3D dimensional Chen system to confuse the pixel positions. Next, Lorenz hyper chaotic system is applied on the permuted pixels to diffuse their values. Thirdly, the output encrypted images from second step are encoded by utilizing DNA coding rules. Fourthly, these two coded images are integrated by performing DNA addition operation to obtain DNA encrypted image. Lastly, the final encrypted image is obtained by adopting DNA decoding rules. The contributions of this work can be summarized as follows:

1. developing bit replacing technique which has the great role in improving the security of the presented method.
2. overcoming the weakness of low dimensional chaotic systems by employing high dimensional and hyper chaotic systems to generating the key stream of extremely big key size space.
3. solving the problems of utilizing only permutation or only diffusion techniques by adopting permutation/diffusion structure to satisfy high security requirements against known attacks.
4. combining three different techniques including bit replacement, DNA coding rules/operations and chaos theory capabilities in aim to enhance the resistance of presented image cryptosystems to common cryptanalyses.

The remaining of the paper is organized as follows: Section 2 introduces the basic theory of the presented approach including bit replacing technique, chaotic systems and DNA mechanism, Section 3 explains the structure of the proposed image cryptosystem. Simulation results and performance analysis are described in Section 4. Section 5 presents the comparative analysis results, and finally Section 6 summarizes the main conclusions and future research directions.

2 Background

In this section, we explain used chaotic systems and DNA encryption rules and their algebraic operations as shown below:

2.1 Chaotic systems

Chaotic systems have been widely utilized in the image cryptography. They are heavily used in the information security because of their excellent properties in comparison with conventional techniques. In general, chaotic systems are used both for data encryption and key production in the design of cryptographic schemes. However, usage of low dimensional chaotic systems cannot assure sufficient protection for the secret data due to the little key space and easy implementation and simple structure. Thus, implementing chaotic system of higher dimension or hyper-chaotic system [5, 7, 11, 63] is the best choice. Chen and hyper Lorenz chaotic systems are used in our encryption scheme and their details are described briefly as follows.

2.1.1 Chen system

Three dimensional Chen chaotic system which was first introduced in 1999 by G. Chen can be modeled as:

$$\begin{aligned}\dot{x} &= a(y-x) \\ \dot{y} &= (c-a)x - xz + cy \\ \dot{z} &= xy - bz\end{aligned}\quad (1)$$

where a, b, c indicate to the Chen system parameters. In order to be in chaotic state, these parameters should have the following values: $a = 35, b = 3, c \in (20, 28.4)$ [35].

2.1.2 Lorenz's hyper-chaotic system

Four dimensional hyper-chaotic Lorenz system can be expressed as follows:

$$\begin{aligned}\dot{x} &= a(y-x) \\ \dot{y} &= cx - xz - y + ew \\ \dot{z} &= x^4 + y^4 - bz \\ \dot{w} &= -dw\end{aligned}\quad (2)$$

where a, b, c, d, e point to the system parameters. 4D Lorenz hyper system is considered chaotic and can generate chaotic streams when the values of these parameters are: $a = 10, b = 8/3, c = 46, d = 2, e = 12$ [20]. Algorithms 1 and 2 show the pseudo code of generating Chen and Lorenz's hyper-chaotic systems.

Algorithm 1: Generating 3D Chen chaotic system.

```
// Input secret keys:  $x_0, y_0, z_0, a, b, c$ .
// Input step size:  $h$ .
// Input number of iterations:  $n$ .
// Output chaotic sequences:  $x, y, z$ .
1: Let  $x_0 = 0.5, y_0 = -0.5, z_0 = 0.7, a = 35, b = 3, c = 20$ .
2: Let  $h = 0.01$ .
3: Let  $n = 100$ .
4: for  $i=1$  to  $n$  do
5:  $x(i+1) = x(i) + h \times a \times (y(i) - x(i))$ 
6:  $y(i+1) = y(i) + h \times ((c-a) \times x(i) - x(i) \times z(i) + c \times y(i))$ 
7:  $z(i+1) = z(i) + h \times (x(i) \times y(i) - b \times z(i))$ 
8: end for
```

Algorithm 2: Generating 4D hyper-chaotic Lorenz system.

```

// Input secret keys:  $x_0, y_0, z_0, w_0, a, b, c, d, e$ .
// Input step size:  $h$ .
// Input number of iterations:  $n$ .
// Output chaotic sequences:  $x, y, z, w$ .
1: Let  $x_0 = 0.1, y_0 = -0.1, z_0 = 0.2, w_0 = 0.1, a = 10, b = \frac{8}{3}, c = 46, d = 2, e = 12$ .
2: Let  $h = 0.01$ .
3: Let  $n = 100$ .
4: for  $i=1$  to  $n$  do
5:  $x(i+1) = x(i) + h \times a \times (y(i) - x(i))$ 
6:  $y(i+1) = y(i) + h \times (c \times x(i) - x(i) \times z(i) - y(i) + e \times w(i))$ 
7:  $z(i+1) = z(i) + h \times ((x(i).^4) + (y(i).^4) - (b \times z(i)))$ 
8:  $w(i+1) = w(i) + h \times (-d \times y(i))$ 
8: end for

```

2.2 DNA encryption rules and algebraic operations

Generally, sequence of DNA includes 4 bases of nucleic acid: Cytosine (C), Guanine (G), Thymine (T) and Adenine (A). C joins with G and A joins T, or in other words, (C, G) and (A, T) represent complementary pairs. Since zero and one in binary system structure are complementary, (01, 10) and (00, 11) are also complementary. Twenty-four sorts of encryption rules can be found if C, G, A and T are employed to code 01, 10, 00 and 11, but only eight sorts of them can fulfill the complement rules of Watson Crick. These rules are given in Table 1. The value of each pixel in grayscale image with binary length of 8 can be represented as DNA code with length of 4. For instance, if the decimal pixel value in grayscale image is 170, then the corresponding binary string to this pixel is 101,011, if Rule 2 in Table 1 is adopted; the obtained DNA coded sequence is CCCT. Contrariwise, the coded sequence (CCCT) can be decoded by utilizing the same rule to restore the original binary string (10101011). But if Rule 3 is employed to decode the DNA coded sequence (CCCT), then the gained binary string is 00000010 which corresponds to the number 2 in decimal system.

Algebraic processes for DNA encryption are described by researchers so as to simplify the DNA implementation in cryptography like addition and subtraction processes. These two processes correspond to the addition and subtraction operations of binary system. Eight rules for each DNA addition and DNA subtraction processes are existed according to eight DNA sorts of encryption methods. DNA addition and subtraction processes are provided in Table 2 [13, 55, 56]. Rule 2 is exploited in this scheme for image pixels encoding/decoding. Also, DNA addition operation rules are used in this scheme for image encryption because as seen from Table 2, the outcomes

Table 1 Encoding rules of DNA

+	A	G	C	T	−	A	G	C	T
A	A	G	C	T	A	A	T	C	G
G	G	C	T	A	G	G	A	T	C
C	C	T	A	G	C	C	G	A	T
T	T	A	G	C	T	T	C	G	A

Table 2 Addition and subtraction processes of DNA

DNA bases	Rule 1	Rule 2	Rule 3	Rule 4	Rule 5	Rule 6	Rule 7	Rule 8
A	00	00	01	01	10	10	11	11
C	01	10	00	11	00	11	01	10
G	10	01	11	00	11	00	10	01
T	11	11	10	10	01	01	00	00

of this operation in all rows and columns are distinguishing. The details for DNA coding and decoding rules are given in Algorithms 3 and 4.

Algorithm 3: DNA coding rules (Rule 2).

```

// Input decimal number: S.
// Output DNA coded sequence: H
1: B=decToBinary(S)                                % B of size (1 × 8)
2: H=zeros(1, 8)
3: L=length(B)
4: for i=1 to L do
5: if B(i) == 0 & B(i + 1) == 0 then
6: H(i)=A
7: else if B(i) == 1 & B(i + 1) == 0 then
8: H(i)=C
9: else if B(i) == 0 & B(i + 1) == 1 then
10: H(i)=G
11: else if B(i) == 1 & B(i + 1) == 1 then
12: H(i)=T                                % H of size (1 × 4)
13: end if
14: end for

```

Algorithm 4: DNA decoding rules (Rule 2).

```

// Input DNA coded sequence: H.
// Output decimal number: S.
1: D=zeros(1, 8)
2: L=length(H)                                % H of size (1 × 4)
3: for i=1 to L do
4: if H(i) == A then
5: D(i)=00
6: else if H(i) == C then
7: D(i)=10
8: else if H(i) == G then
9: D(i)=01
10: else if H(i) == T then
11: D(i)=11                                % D of size (1 × 8)
12: end if
13: end for
14: S=BinaryTodec(D)

```

3 Proposed method

Architecture of the presented image security method is illustrated in Figs. 2 and 3, respectively. The encryption operation in the suggested method includes six main phases. In the initial phase, the input image is partitioned into two independent images based on bit replacing technique. In the second phase, the pixels in the resultant two images in the previous phase are permuted via random key stream produced by Chen chaotic system so as to lessen the potency relationship between image pixels. Only pixels locations are modified in this phase not pixels

values, hence the histogram of image will stay unchanged. To improving the security, diffusion operation is utilized on the permuted images via random key stream generated by hyper chaotic Lorenz system to vary the pixels values in the third phase. Image histogram in this stage is completely different from the input image histogram thereby improves the system resistance significantly against statistical analysis attacks. Next, to further encrypt the diffused images and enhance the security, the diffused images acquired in the last step are transformed to coded sequences in the fourth phase based on DNA encryption rules. In the fifth phase, the encoded DNA images from fourth step are integrated together by adopting DNA addition process to produce the encrypted DNA image. This encrypted DNA image is decoded in the final step by exploiting DNA decryption rules to get the eventual encrypted image. The encryption and decryption algorithms which are shown in the Figs. 2 and 3, respectively are outlined in the following subsections.

3.1 Bit replacing technique (BRT)

Bit replacing is a new proposed technique in this cryptosystem. It increases the difficulty of the cryptanalysis of the input image during transmission by partitioning it into two different images. Initially, the original image is transformed to binary image. It means that each pixel in the image is converted to its corresponding binary stream that consists of 0 s and 1 s bits. For example, the 97 pixel in the 2×2 image as illustrated in the Fig. 1, it is converted to its binary equivalent

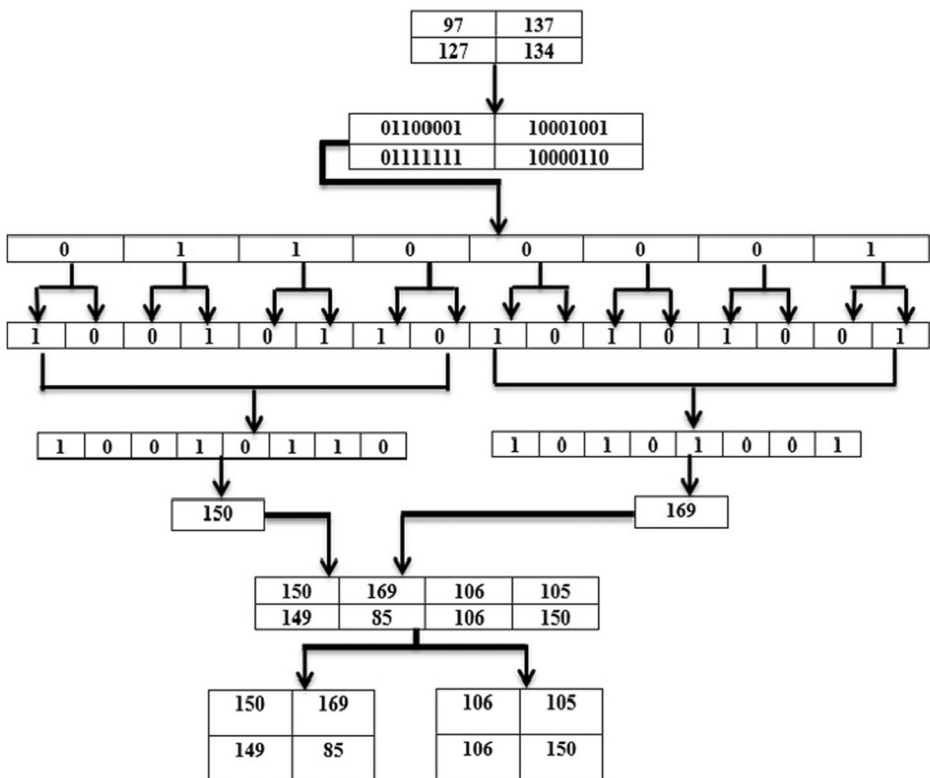


Fig. 1 Block diagram of bit replacing technique

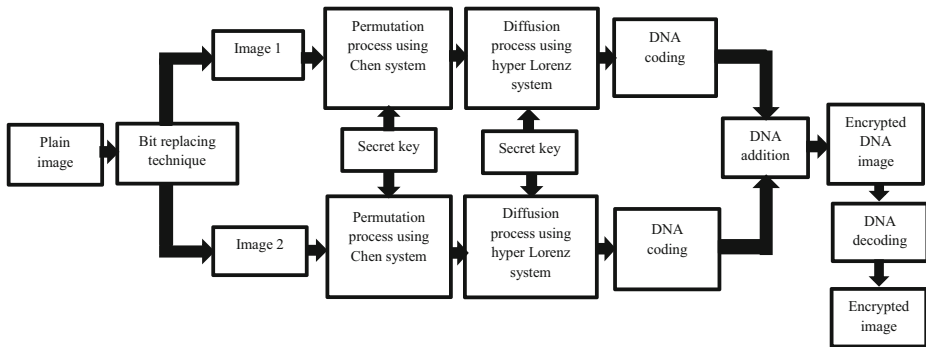


Fig. 2 Block diagram of the proposed image encryption method

string (01100001) and then the first pixel bit (0) is replaced by bits (10) and the second bit (1) is replaced by bits (01). This operation yields 16 binary bits instead of 8 bits. Next, the consequent 16 bits are separated into two equal parts, then each part is transformed back to its corresponding decimal form. The first part gives the pixel value of 150, while the second part gives the pixel value of 169. Thus, pixel value 97 ends up with two different values 150 and 169. This strategy is continued for the remaining pixels 137, 127 and 134 to obtain two different output pixel values for each input pixel. As a result, the input 2×2 image is transformed into 2×4 output image. The final stage is separating the resultant image matrix into two equal matrices with size 2×2 . Adopting this new technique in the proposed cryptosystem can highly improve the efficiency and security of the suggested method, algorithm 5 pseudo code below explains the proposed bit replacing technique with more details.

Algorithm 5: The proposed bit replacing technique.

```

// A: Input image of size (2 × 2).
// H1 and H2: Output images each of size (2 × 2).
1: [m, n] = size(A)
2: B = decToBinary(A) % B of size (4 × 8)
3: C = reshape(B, (1, m × n × 8)) % C of size (1 × 32)
4: D = zeros(1, m × n × 8)
5: L = length(C) % H = 32
6: for i = 1 to L do
7: if C(i) == 0 then
8: D(i) = 10 % D of size (1 × 64)
9: else if C(i) == 1 then
10: D(i) = 01
11: end if
12: end for
13: E = reshape(D, (m × n × 8 × 2, 1)) % E of size (64 × 1)
14: F = reshape(E, (m × n, 16)) % F of size (4 × 16)
15: F1 = F(:, 1 to 16/2) % F1 of size (4 × 8)
16: F2 = F(:, 16/2 + 1 to 16) % F2 of size (4 × 8)
17: G1 = BinaryToDec(F1) % G1 of size (4 × 1)
18: G2 = BinaryToDec(F2) % G2 of size (4 × 1)
19: H1 = reshape(G1, (m, n)) % H1 of size (2 × 2)
20: H2 = reshape(G2, (m, n)) % H2 of size (2 × 2)

```

3.2 Encryption procedure

The image encryption procedure is described as below:

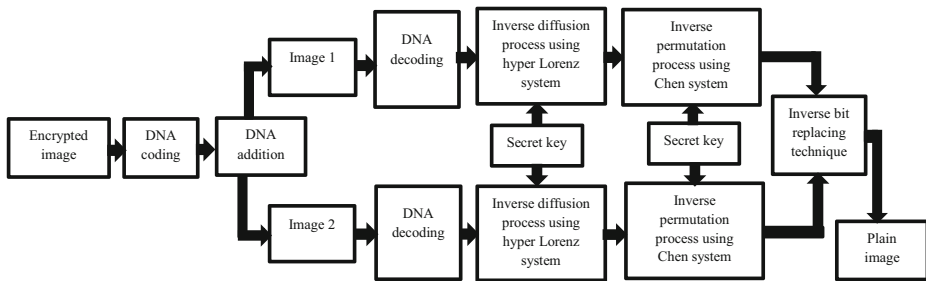


Fig. 3 Block diagram of the proposed image decryption method

- Step 1: Transform the input gray image A (M, N) from decimal form to binary form to generate the equivalent binary matrix B ($M \times N, 8$).
- Step 2: Apply bit replacing technique according to (Section 2.1) on the matrix B to get the new matrix S ($M \times N, 16$).
- Step 3: Split the resultant matrix S from Step 2 into two equal matrices $N_1(M \times N, 8)$ and $N_2(M \times N, 8)$. Then, N_1 and N_2 are converted to their equivalent decimal matrices $U_1(M, N)$ and $U_2(M, N)$.
- Step 4: Produce three random streams $X_1 = \{x_1, x_2, \dots, x_{M \times N}\}$, $Y_1 = \{y_1, y_2, \dots, y_{M \times N}\}$, $Z_1 = \{z_1, z_2, \dots, z_{M \times N}\}$ from Chen chaotic system in (Section 2.2.1) based on its parameters and the initial values (a, b, c, x_0, y_0, z_0).
- Step 5: Generate the key streams X_2, Y_2, Z_2 by processing X_1, Y_1, Z_1 as seen in the formulas:

$$\begin{aligned} X_2 &= \text{mod}(\text{floor}(X_1 \times 10^{14}), 256) \\ Y_2 &= \text{mod}(\text{floor}(Y_1 \times 10^{14}), 256) \\ Z_2 &= \text{mod}(\text{floor}(Z_1 \times 10^{14}), 256) \end{aligned} \quad (3)$$

Then, X_2 sequence is chosen to shuffle the matrices U_1 and U_2 . Next, X_2 stream is reshaped as U_1 or U_2 size to gain the initial permutation matrix L (M, N) according to the equation:

$$L = \text{reshape}(X_2, M, N) \quad (4)$$

- Step 6: Generate permutation matrix L' by sorting L in ascending order as follows:

$$[L', D] = \text{sort}(L) \quad (5)$$

where D refers to the index of sequence and sort indicates the sort function. This operation is clarified in Fig. 4.

- Step 7: Scramble the pixels locations of rows and columns in the matrices U_1 and U_2 by utilizing the permutation chaotic matrix L' to acquire the permuted images $E_1(M, N)$ and $E_2(M, N)$ as illustrated below:

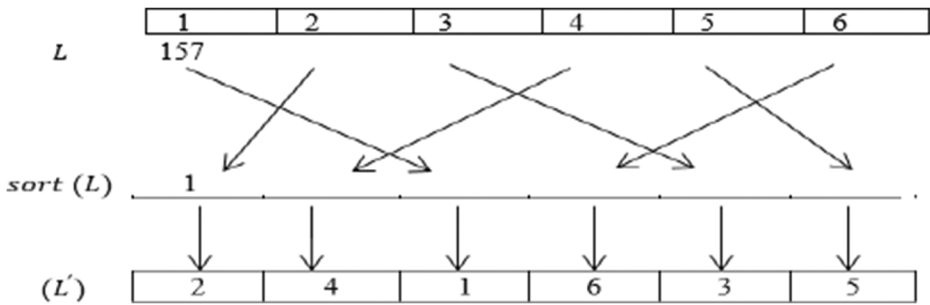


Fig. 4 Example of generating the permutation matrix

$$\begin{aligned} E_1(i, j) &= U_1(L'(i), L'(j)) \\ E_2(i, j) &= U_2(L'(i), L'(j)) \end{aligned} \quad (6)$$

Step 8: Produce four random streams $X'_1 = \{x'_1, x'_2, \dots, x'_{M \times N}\}$, $Y'_1 = \{y'_1, y'_2, \dots, y'_{M \times N}\}$, $Z'_1 = \{z'_1, z'_2, \dots, z'_{M \times N}\}$, $W'_1 = \{w'_1, w'_2, \dots, w'_{M \times N}\}$ from Lorenz hyper chaotic system in (Section 2.2.2) based on its parameters and initial values $(a, b, c, d, e, x_0, y_0, z_0, w_0)$.

Step 9: Generate the key stream X'_2, Y'_2, Z'_2, W'_2 by processing X'_1, Y'_1, Z'_1, W'_1 as defined in the equations:

$$\begin{aligned} X'_2 &= \text{mod} \left(\text{floor} \left(X'_1 \times 10^{14} \right), 256 \right) \\ Y'_2 &= \text{mod} \left(\text{floor} \left(Y'_1 \times 10^{14} \right), 256 \right) \\ Z'_2 &= \text{mod} \left(\text{floor} \left(Z'_1 \times 10^{14} \right), 256 \right) \\ W'_2 &= \text{mod} \left(\text{floor} \left(W'_1 \times 10^{14} \right), 256 \right) \end{aligned} \quad (7)$$

Then, Y'_2 sequence is chosen to diffuse the matrices E_1 and E_2 . Next, Y'_2 stream is reshaped as E_1 or E_2 size to get the diffusion matrix $Y_3 (M, N)$ according to the formula:

$$Y_3 = \text{reshape} \left(Y'_2, M, N \right) \quad (8)$$

Step 10: Encrypt the pixel values of rows and columns in the permuted matrices E_1 and E_2 by applying the diffusion chaotic matrix Y_3 to obtain the diffused images $E'_1(M, N)$ and $E'_2(M, N)$ as follows:

$$\begin{aligned} E'_1(i, j) &= E'_1(i, j-1) \oplus E_1(i, j) \oplus Y_3(i, j) \\ E'_2(i, j) &= E'_2(i, j-1) \oplus E_2(i, j) \oplus Y_3(i, j) \end{aligned} \quad (9)$$

- Step 11: Transform the decimal diffused matrices E'_1 and E'_2 into their corresponding binary matrices $E_3(M \times N, 8)$ and $E_4(M \times N, 8)$.
- Step 12: Implement Rule 2 according to DNA encryption rules in Table 1 to encode E_3 and E_4 , the acquired coded matrices are $C_1(M \times N, 4)$ and $C_2(M \times N, 4)$.

$$\begin{aligned} C_1 &= \text{Encode}(E_3) \\ C_2 &= \text{Encode}(E_4) \end{aligned} \quad (10)$$

- Step 13: Perform DNA addition process according to Table 2 between encoded matrices C_1 and C_2 to produce the concatenated matrix $S_1(M \times N, 4)$ via the formula:

Algorithm 6: The proposed image encryption method.

```
// A: Input image of size (M, N).
// H: Output image of size (M, N).
1: [M, N]=size(A)
2: B=decToBinary(A)                                % B of size (M × N, 8)
3: Apply bit replacing technique (Algorithm 1) on B to generate S    % S of size (M × N, 16)
4: N1=S(:, 1 to  $\frac{16}{2}$ )                                % N1 of size (M × N, 8)
5: N2=S(:,  $\frac{16}{2} + 1$  to 16)                            % N2 of size (M × N, 8)
6: U1=BinaryTodec(N1)                                % U1 of size (M, N)
7: U2=BinaryTodec(N2)                                % U2 of size (M, N)
8: Produce X1, Y1, Z1 according to Algorithm 2.
9: Generate X2, Y2, Z2 from X1, Y1, Z1 according to Eq. (3).
10: Reshape X2 to gain L according to Eq. (4).                % L of size (M, N)
11: Sorting L in ascending order to gain L' according to Eq. (5).
12: Scramble U1, U2 via L' to get E1, E2 (permutation) according to Eq. (6).
13: Produce X'1, Y'1, Z'1, W'1 according to Algorithm 3.
14: Generate X'2, Y'2, Z'2, W'2 from X'1, Y'1, Z'1, W'1 according to Eq. (7).
15: Reshape Y'2 to gain Y3 according to Eq. (8).            % Y3 of size (M, N)
16: Apply Y3 on E1, E2 to get E'1, E'2 (diffusion) according to Eq. (9).
17: E3=decToBinary(E'1)                                % E3, E4 of size (M × N, 8)
18: E4=decToBinary(E'2)
19: Encode E3, E4 to get C1, C2 according to Eq. (10).      % E3, E4 of size (M × N, 4)
20: Add C1, C2 to get S1 according to Eq. (11).
21: Decode S1 to obtain S2 according to Eq. (12).          % S2 of size (M × N, 8)
22: H=BinaryTodec(S2)                                % H of size (M, N)
```

$$S_1(i, j) = C_1(i, j) + C_2(i, j) \quad (11)$$

- Step 14: Implement Rule 2 according to DNA decryption rules in Table 1 to decode S_1 matrix, the resultant binary matrix is $S_2(M \times N, 8)$.

$$S_2 = \text{Decode}(S_1) \quad (12)$$

Step 15: Convert the binary matrix S_2 to its equivalent decimal matrix H (M , N), H represents the output encrypted image.

If the input image is a color image, A (M , N , 3), then A is divided to three matrices R (M , N), G (M , N) and B (M , N). After that, Steps (1–15) are executed on R , G and B separately to earn three encrypted images. The definitive step is joining them to produce the eventual cipher image H (M , N , 3), algorithm 6 and Fig. 5 are presented to give more details on the proposed encryption steps.

3.3 Decryption procedure

In the decryption process, the above encryption steps are carried out in reverse order. It means that the input image to the decryption algorithm is H (encrypted image), while the output image is A (original image). The same keys ((a, b, c, x_0, y_0, z_0)) of

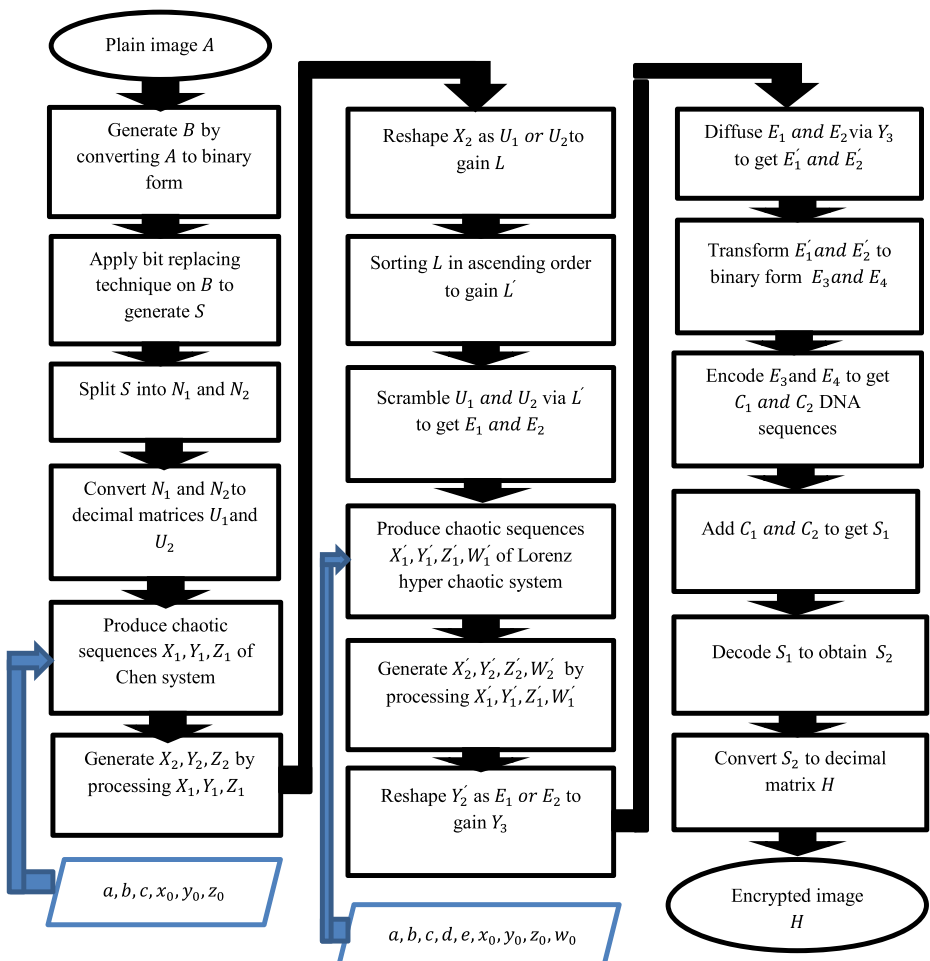


Fig. 5 Flow chart of the proposed image encryption method

Chen system and $(a, b, c, d, e, x_0, y_0, z_0, w_0)$ of Lorenz hyper chaotic system) employed in the encryption procedure are required for decryption procedure. The keys must be conserved safely and transferred to the recipient who can restore the input plain image from the output encrypted image by employing those secret keys in the decryption operation.

3.4 Case study (an example of the proposed method)

Examples of the encryption and decryption processes in the introduced technique are exhibited in Figs. 6 and 7, respectively. In Fig. 6, the input image is resized to (3×3) in order to understand the proposed encryption process. First, the plain image is divided into two different images (Image 1 and Image 2) according to Bit Replacing Technique (BRT) in Section 2.1. In the next step, Images 1 and 2 are scrambled by changing their pixels locations according to chaotic numbers generated from Chen system (Eq. 1). After that, the resultant scrambled images are diffused by changing their values according to chaotic values produced from Lorenz's hyper system (Eq. 2). Then, the diffused images are encoded by employing Rule 2 (Table 1). The last two pixels (170 and 206) are taken from the diffused matrices as example to illustrate this encoding operation. 170 and 206 are converted from their decimal form to binary form: 10101010 and 11,001,110, respectively, these sequences are encoded to obtain CCCC and TATC sequences, respectively by carrying out DNA computing rules. The coded sequences are then added together by utilizing DNA addition operation to gain the encoded sequence GCGA. Next, this encoded sequence is decoded via DNA decoding rules (Rule 2) to acquire the binary sequence: 01100100 which is equivalent to decimal number 100. The above

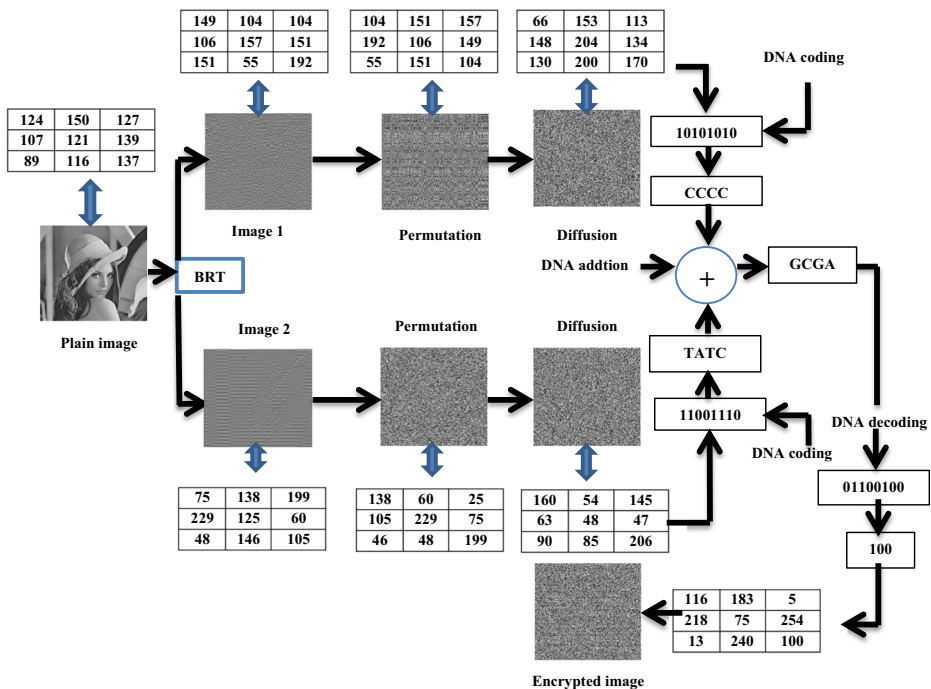


Fig. 6 An example of the encryption process in the proposed technique

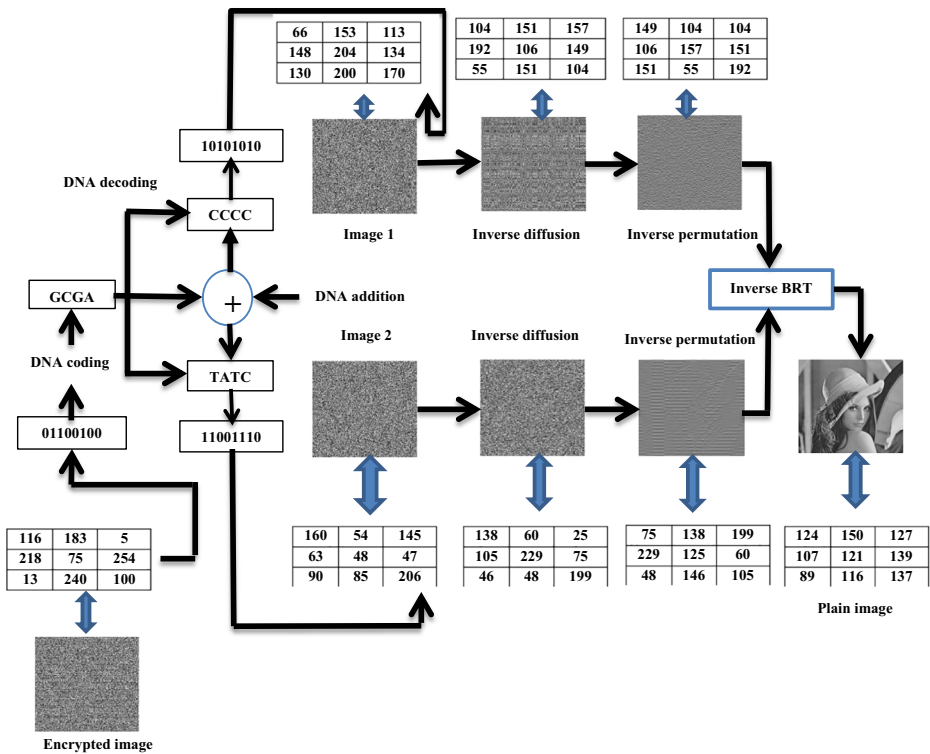


Fig. 7 An example of the decryption process in the proposed technique

steps are applied on all pixels in the diffused images (Image 1 and Images 2) to get the final cipher image. The decryption process in Fig. 7 is similar to encryption process in Fig. 6 but in adverse order. As seen from Fig. 6, the adopted new technique (Bit Replacing Technique) can produce images that are look like chaotic images and they are totally different from the original image. After applying the chaotic systems and DNA encoding/addition rules on the resultant images, the eventual image became more noisy and unlike the input image. As a result, the attacker can not acquire any useful information from the consequent output image thereby the presented scheme can defeat several sorts of attacks and analyses as seen in the following subsections.

4 Experimental simulation results analysis

The simulation results are presented in this section to clarify the performance, sturdiness and randomness of the proposed image cryptosystem. Gray and color images are utilized in these tests to examine the suggested method. The sample images are extracted from SIPI image database with TIFF format, each of size 512×512 . Moreover, three other images are chosen with JPEG format and size of 256×256 to assess the presented method. The images that employed in our experiments are shown in the Fig. 8. Personal laptop with 4GB memory, Intel Core i3, 2.40 GHz processor, Windows 7, and MATLAB 2013a software is utilized to implement all encryption and decryption experiments of the presented method. The secret



Fig. 8 Test images **a** Gray Lena, **b** Gray Baboon, **c** Gray Peppers, **d** Boat, **e** Color Lena, **f** Color Baboon, **g** Color Peppers, **h** Barbara, **i** Braine, **j** Fingerprint and **k** City

keys for Chen system are appointed as: $x_0 = 0.5$, $y_0 = -0.5$, $z_0 = 0.7$, $a = 35$, $b = 3$, $c = 20$, whereas the secret keys for Lorenz hyper system are designated as: $x_0 = 0.1$, $y_0 = -0.1$, $z_0 = 0.2$, $w_0 = 0.1$, $a = 10$, $b = 8/3$, $c = 46$, $d = 2$, $e = 12$. The encryption and decryption results of gray Baboon, color Baboon, Braine, Fingerprint, and city images are shown in the Fig. 9. Figure 9a–e imply that the output encrypted images produced by our method are chaotic

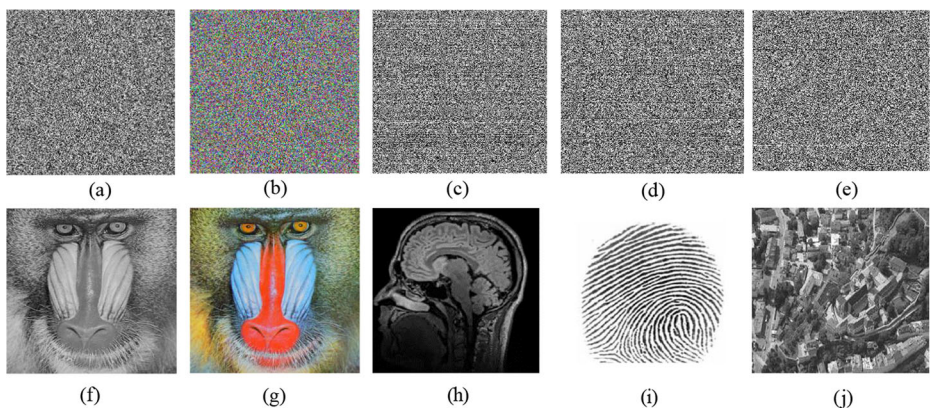


Fig. 9 Encryption and decryption outcomes **a** Encrypted gray Baboon, **b** Encrypted color Baboon, **c** Encrypted Braine, **d** Encrypted Fingerprint, **e** Encrypted City, **f** Decrypted image of (a), **g** Decrypted image of (b), **h** Decrypted image of (c), **i** Decrypted image of (d), **j** Decrypted image of (e)

images thereby no important data about their input versions can be extracted from them. On the other hand, Fig. 9f–j indicate that the decrypted images and their corresponding plain versions are identical. The plain images are easily restored by utilizing right parameters and initial conditions values. In the following subsections, the method performance analysis results are presented as follows:

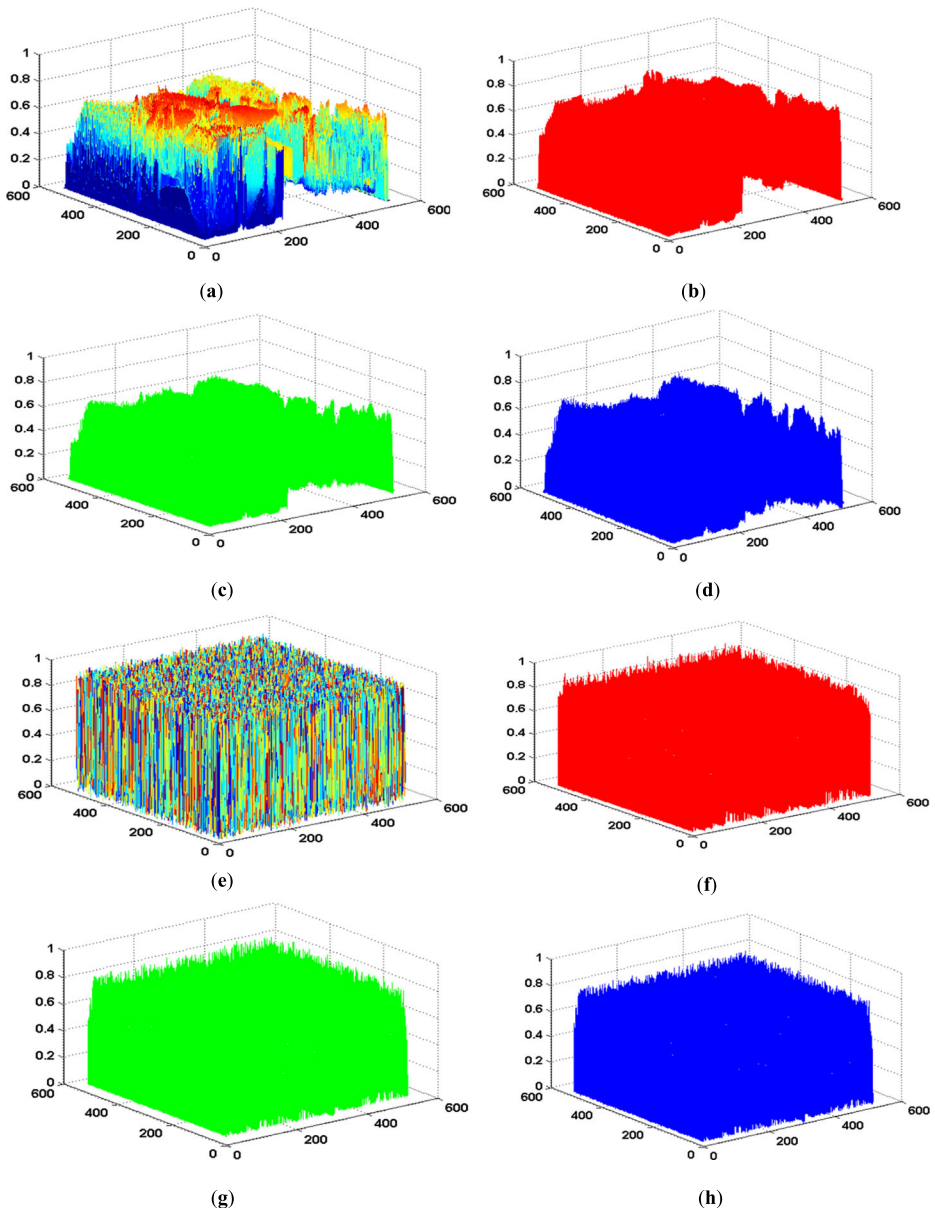


Fig. 10 Spatial distribution analysis in the **a** RGB channel, **b** Red, **c** Green, **d** Blue channels of the plain Barbara image, **e** RGB channel, **f** Red, **g** Green, **h** Blue channels of the encrypted Barbara image

4.1 Spatial distribution analysis

In this analysis, each image pixel is represented in two dimensions, x -axis and y -axis, whilst color intensity of pixel is represented in z -axis [27, 53]. The pixels distribution in spatial domain for the plain and encrypted Barbara image in three color channels (red, green and Blue) is illustrated in Fig. 10. It is evident from Fig. 10a-d that there are several valleys and peaks because the plain image contains a lot of information, and the pixels of input image in many locations are distributed unevenly in the three channels, whereas this distribution is very uniform in the encrypted image and the pixels are filled between (0:255) space according to Fig. 10e-h for the three channels. Thus, it is difficult for the adversary to access or get any

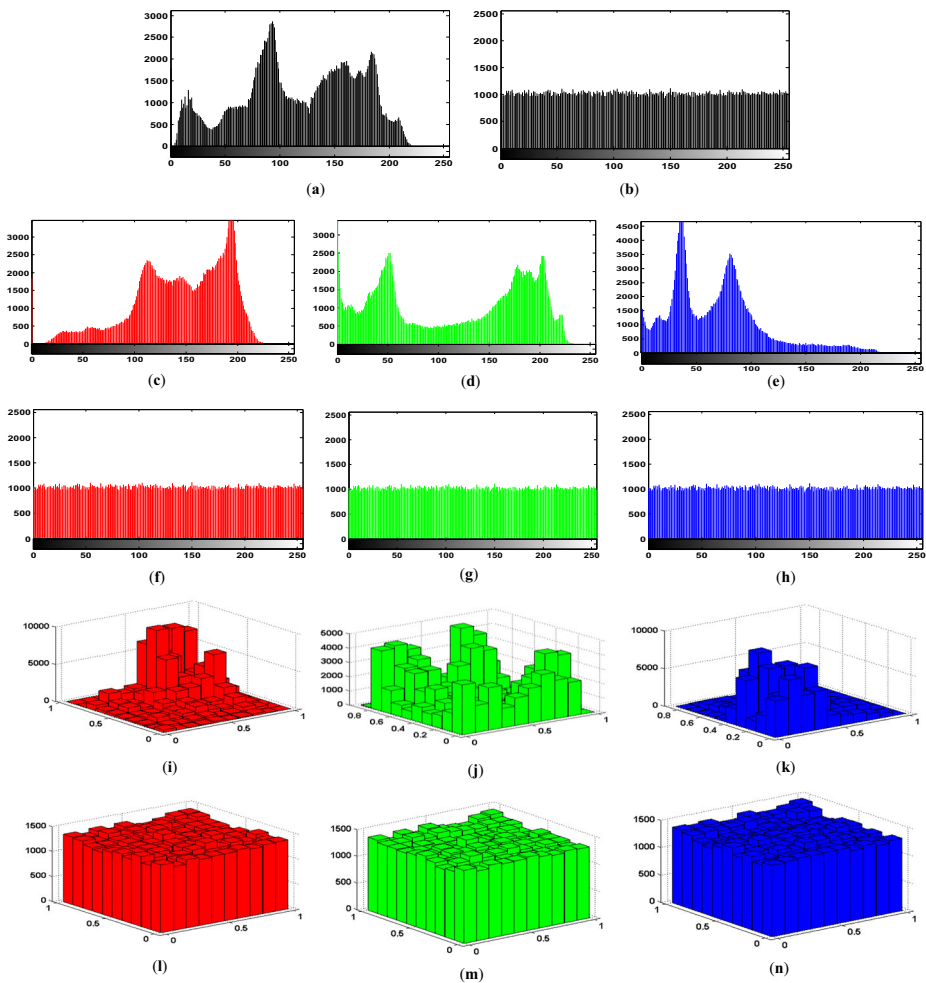


Fig. 11 Histogram analysis **a** Histogram of plain gray image, **b** Histogram of encrypted gray image, **c-e** Histogram of plain color image for RGB components, **f-h** Histogram of encrypted color image for RGB components, **j-k** 3D histogram of plain color image for RGB components, **l-n** 3D histogram of encrypted color image for RGB components

valuable information from the uniform distribution of cipher image by the statistical analysis attack.

4.2 Histogram analysis

Histogram plot exhibits the pixel density in the image. In general, the input image histogram is not uniform, while the encrypted image histogram in ideal cryptosystem should be uniform to resist statistical analysis attack [29]. The histograms of input and output images for gray and color Peppers image are given in Fig. 11. According to the scores in Fig. 11a, c–e, it can be seen that the pixel values in the plain image are grouped around some values, whereas the encrypted image histograms in Fig. 11b, f–h possess a uniform distribution and the difference between them is obviously visible. Moreover, 3D histogram for color image Peppers in the RGB components is illustrated in Fig. 11. It is easily to notice that the encrypted image histograms in Fig. 11i–n are equally uniform and they are essentially different from plain image histograms in Fig. 11i–k. The histogram uniformity of the encrypted image indicates the strength of the presented scheme in modifying the pixels of the plain image. This manifests that the proposed encryption scheme can endure efficiently statistical analysis. However, Chi-square measure which is given in the formula below is employed to assure histogram uniformity [46]:

$$\chi^2 = \sum_{i=1}^{256} \frac{(o_i - e_i)^2}{e_i} \quad (13)$$

where o_i , e_i symbolize to the obtained and expected frequencies, respectively. The ideal scores for this measure should be smaller than 293.25 for freedom degree ($d = 255$) and significance level ($\alpha = 0.05$). The χ^2 outcomes for the encrypted test images acquired via the proposed approach are provided in Table 3. This table reveals that χ^2 values for all encrypted images are lower than its optimal value, which denotes that the output image possesses a uniform histogram, hence the suggested mechanism passed this test. Furthermore, another test which is related to Chi-square test called color frequency test is applied also to measure the probability of pixels distribution before and after the encryption process. This test is given by [9]:

Table 3 Results of entropy, Chi-square and color frequency test measures between plain and encrypted images

Images	Plain Image entropy	Encrypted image entropy (Global)	Encrypted image entropy (Local)	χ^2	p plain image	p encrypted image
Gray Lena	7.4455	7.9993	7.9071	240.3730	0	0.99591
Gray Baboon	7.3585	7.9993	7.9066	245.3320	0	0.99064
Gray Peppers	7.5712	7.9993	7.9065	272.1992	0	0.95950
Boat	7.1238	7.9993	7.9063	252.4395	0	0.96388
Color Lena	7.7503	7.9994	7.9062	235.4941	0	0.92636
Color Baboon	7.7626	7.9993	7.9068	262.1758	0	0.93527
Color Peppers	7.6612	7.9993	7.9070	267.7266	0	0.98332
Barbara	7.6986	7.9992	7.9065	272.7305	0	0.96533
Braine	6.4568	7.9968	7.9057	290.2813	0	0.93590
Fingerprint	6.5566	7.9970	7.9063	275.0625	0	0.99825
City	7.4549	7.9969	7.9042	278.2266	0	0.91835

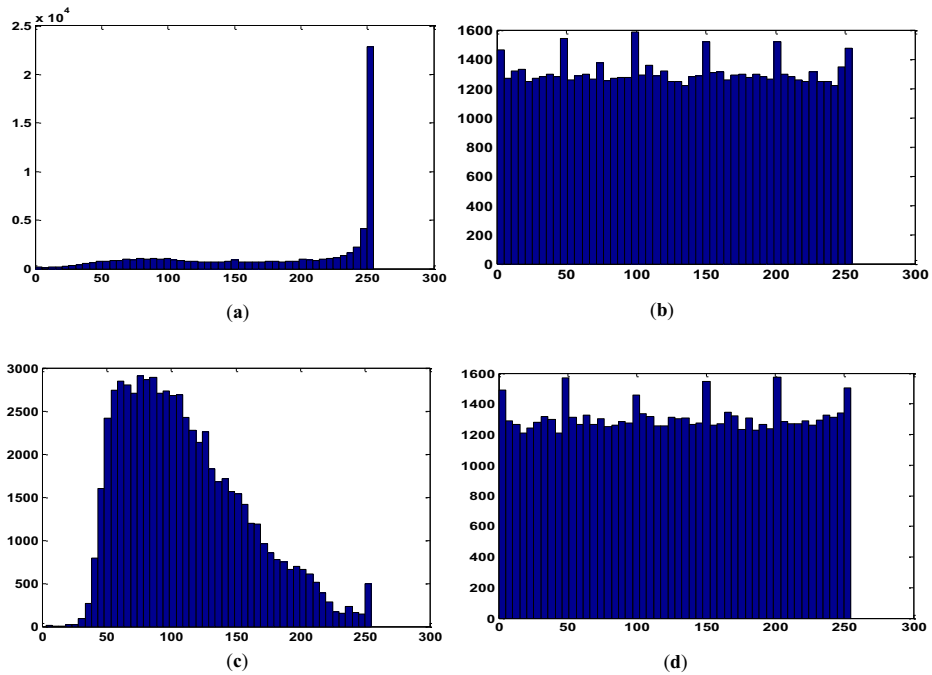


Fig. 12 1st order statistical attack analysis **a** Histogram of plain image (Fingerprint), **b** Histogram of encrypted image (Fingerprint), **c** Histogram of plain image (City), **d** Histogram of e image (City)

$$p = \frac{1}{2^{\frac{\nu}{2}} \Gamma\left(\frac{\nu}{2}\right)} \int_0^{X_{k-1}^2} t^{\frac{\nu}{2}-1} e^{-\frac{t}{2}} dt \quad (14)$$

where ν refers to the freedom degree and $\nu = k - 1$ indicates the categories number of distinct colors. For the input image, the probability distribution (p) should be near to zero, while p value for the encrypted image in ideal cryptosystem should be near to 1. The p results for the plain and encrypted images obtained by the presented approach are displayed in Table 3. The table shows that the p outcomes for the input test images are 0. On the other hand, the p outcomes for the encrypted versions are close to 1. These scores indicate the ability of the proposed image cryptosystem to change the pixel intensity values after applying the encryption process. Thus, the introduced technique can pass also this test.

To measure the resistance of the presented scheme against 1st order statistical attack analysis [39], the histograms of the input test images: Fingerprint and City with JPEG format are plotted in Fig. 12a and c, and their corresponding encrypted histograms are plotted in Fig. 12b and d. The difference between the image histograms before and after the encryption process is very obvious in this figure. It can be deduced that the encryption operation is capable of changing the original image pixels and the proposed method can defeat this attack efficiently.

4.3 Global entropy analysis

The complexity, randomness and uncertainty in the image can be calculated by using global entropy test as follows [49]:

$$H(X) = - \sum_{i=0}^n P(x_i) \log_2 P(x_i) \quad (15)$$

where $H(X)$ refers to the global entropy, $P(x_i)$ points to the likelihood of the data source x_i and for a typical random source, $H(X)$ should be 8. The outcomes for original and ciphered images after applying the introduced approach are listed in Table 3. It can be found that all experimental scores of the output encrypted images in this table are so near to the idealistic entropy value, thus the presented method possesses perfect security to counter entropy analysis.

4.4 Local entropy analysis

The randomness of encrypted image can be measured by employing another important criterion is called local entropy test which is determined as follows [45, 53, 66]:

$$\overline{H_{(k,T_B)}(S)} = \sum_{i=1}^k \frac{H(S_i)}{k} \quad (16)$$

where $\overline{H_{(k,T_B)}(S)}$ represents the local entropy, $S_1, S_2, \dots, S_k$ refer to non-overlapping blocks in the output image that are randomly selected with T_B pixels, whilst $H(S_i)$ indicates the global entropy of the chosen block S_i . If k and T_B are set as 30 and 1936, respectively, then $\overline{H_{(k,T_B)}(S)}$ should be in the interval (7.901901305, 7.903037329). The test outcomes of local entropy for multiple encrypted images are calculated in Table 3. These outcomes point out that $\overline{H_{(k,T_B)}(S)}$ for all sample encrypted images are greater than 7.903 thereby the presented cryptosystem can produce encrypted image with excellent randomness.

4.5 Correlation analysis

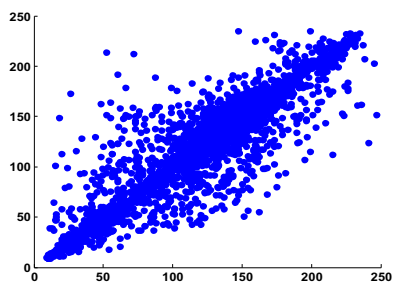
Generally, neighboring pixels in the input image always possess vigorous relation due to the large amount of information in the image. Ideal encryption system should reduce this relationship among pixels as much as possible in order to prevent statistical analysis attack [12, 37]. The following formulas are utilized to calculate the correlation (r_{xy}) between two neighboring pixels in various directions:

$$\begin{aligned} E(x) &= \frac{1}{N} \sum_{i=1}^N x_i \\ D(x) &= \frac{1}{N} \sum_{i=1}^N (x_i - E(x))^2 \\ \text{cov}(x, y) &= \frac{1}{N} \sum_{i=1}^N (x_i - E(x))(y_i - E(y)) \\ r(x, y) &= \frac{\text{cov}(x, y)}{\sqrt{D(x)} \times \sqrt{D(y)}} \end{aligned} \quad (17)$$

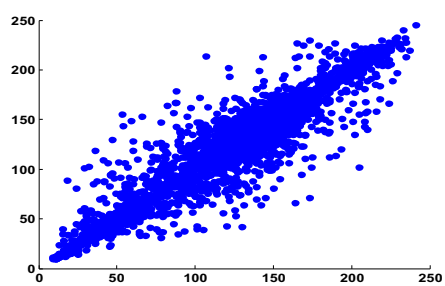
Where (x, y) pair represents the values of two neighboring pixels in the input or output image, $E(x)$ and $E(y)$ symbolize to the expected values of x and y , respectively, $D(x)$ and $D(y)$ refer to the variance of x and y , respectively. In order to assess the presented scheme before and after encryption process, 5000 couples of adjacent pixels are picked out randomly in the experiments from input and output images along horizontal, vertical and diagonal orientations, and

Table 4 Results of correlation coefficients among adjacent pixels in the plain and encrypted images

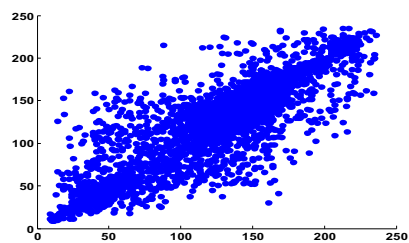
Images	Horizontal		Vertical		Diagonal	
	Plain	Encrypted	Plain	Encrypted	Plain	Encrypted
Gray Lena	0.9728	−0.0215	0.9838	−0.0113	0.9602	0.0089
Gray Baboon	0.8708	0.0141	0.7535	−0.0075	0.7354	0.0027
Gray Peppers	0.9753	0.0041	0.9709	0.000057955	0.9358	0.00097232
Boat	0.9614	−0.0031	0.9771	0.0166	0.9506	−0.0086
Color Lena	0.9795	0.00071219	0.9891	−0.0027	0.9706	−0.0104
Color Baboon	0.9272	−0.0143	0.8526	−0.0174	0.8463	0.0091
Color Peppers	0.9669	0.00027303	0.9591	−0.0055	0.9238	0.0142
Barbara	0.9285	−0.0058	0.9681	−0.0117	0.9092	0.0137
Braine	0.9685	0.00030391	0.9689	0.0223	0.9402	0.0090
Fingerprint	0.8490	0.0023	0.7489	−0.0108	0.7087	0.0226
City	0.8461	−0.0083	0.8902	−0.0162	0.7652	0.00061465



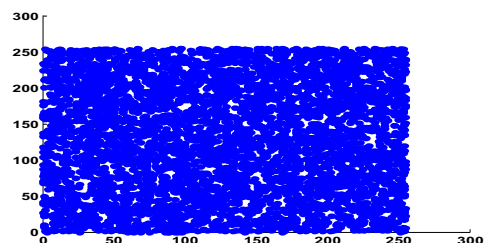
(a)



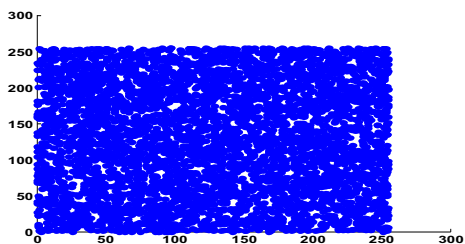
(b)



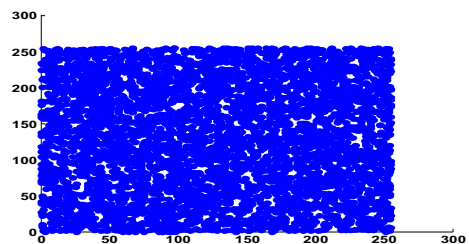
(c)



(d)



(e)



(f)

Fig. 13 Correlation Analysis in **a** Horizontal, **b** Vertical, **c** Diagonal directions of plain Boat image, **d** Horizontal, **e** Vertical, **f** Diagonal directions of encrypted Boat image

the corresponding computed results of correlation coefficient for the plain and encrypted images are shown in the Table 4. The scores in this table reveal that the correlation coefficient values among neighboring pixels in the input images are extremely large, while these values become approximately zero in the corresponding output images along the three orientations. Also, Fig. 13 clarifies the correlation distribution in three orientations for input image Boat and its corresponding output encrypted image. It can be observed from Fig. 13a-c that the neighboring pixel distributions in the plain image are extremely concentrated due to the potent correlation existed in the plain image. Conversely, the adjacent pixel distributions in the encrypted image are evenly distributed, which manifests that the output image possesses negligible correlation. Table 4 and Fig. 13 reveal that the presented technique removed the existing relation among neighboring pixels in the input image. Hence after implementing the encryption process proposed by this method, the encrypted image can resist statistical data analysis.

4.6 Signal to noise ratio (SNR)

SNR criteria is utilized to demonstrate the difference between the input image (X) and output encrypted image (Y). This measure is computed as follows [30]:

$$SNR = \frac{\sum_{i=1}^M \sum_{j=1}^N [X(i, j)]^2}{\sum_{i=1}^M \sum_{j=1}^N [X(i, j) - Y(i, j)]^2} \quad (18)$$

4.7 Mean square error (MSE) and peak signal to noise ratio (PSNR)

The quality between the plain and encrypted images is measured via MSE and PSNR. These two parameters are described as follows [50]:

$$MSE = \frac{1}{M \times N} \sum_{i=1}^M \sum_{j=1}^N [X(i, j) - Y(i, j)]^2 \quad (19)$$

$$PSNR = 10 \times \log_{10} \left[\frac{255 \times 255}{MSE} \right] \quad (20)$$

4.8 Structural similarity index measure (SSIM)

SSIM is used to determine the likeness between the input image (I_1) and output image (I_2). The SSIM is given as [44]:

$$SSIM(I_1, I_2) = \frac{(2\mu_{I_1 I_2} + \alpha)(2\sigma_{I_1 I_2} + \beta)}{(\mu_{I_1}^2 + \mu_{I_2}^2 + \alpha)(\sigma_{I_1}^2 + \sigma_{I_2}^2 + \beta)} \quad (21)$$

where μ , σ^2 , and σ refer to the mean, variance and covariance, respectively between I_1 and I_2 , whilst α and β represent the constants.

Table 5 Results of SNR, MSE, PSNR, SSIM, MAE and RMSE Metrics between the Plain and Encrypted Images

Images	SNR (dB)	MSE	PSNR (dB)	SSIM	MAE	RMSE
Gray Lena	3.5545	7797.7	9.2111	0.0350	73.1986	88.3047
Gray Baboon	4.0993	7234.2	9.5369	0.0339	70.8518	85.0539
Gray Peppers	2.8991	8472.7	8.8506	0.0333	75.7536	92.0474
Boat	4.1078	8259.4	8.9613	0.0324	74.9673	90.8812
Color Lena	5.1582	10,637	7.8625	0.0331	84.2279	103.1376
Color Baboon	4.0527	8635.6	8.7679	0.0358	76.4112	92.9280
Color Peppers	4.6372	7869.4	9.1714	0.0415	73.4397	88.7094
Barbara	3.7307	8516.2	8.8284	0.0306	75.9600	92.2830
Braine	4.8700	14,356	6.5605	0.0039	98.7437	119.8168
Fingerprint	5.0370	14,450	6.5323	0.0091	99.1857	120.2064
City	2.6246	8008.8	9.0951	0.0080	73.8855	89.4920

4.9 Mean absolute error (MAE)

The difference between the plain and encrypted images is tested by utilizing MAE index, which can be defined as follows [29]:

$$MAE = \frac{1}{M \times N} \sum_{i=1}^M \sum_{j=1}^N |X(i, j) - Y(i, j)| \quad (22)$$

4.10 Root mean squared error (RMSE)

This criterion measures the error between X (the original image) and Y (the encrypted image). RMSE is calculated as follows [64]:

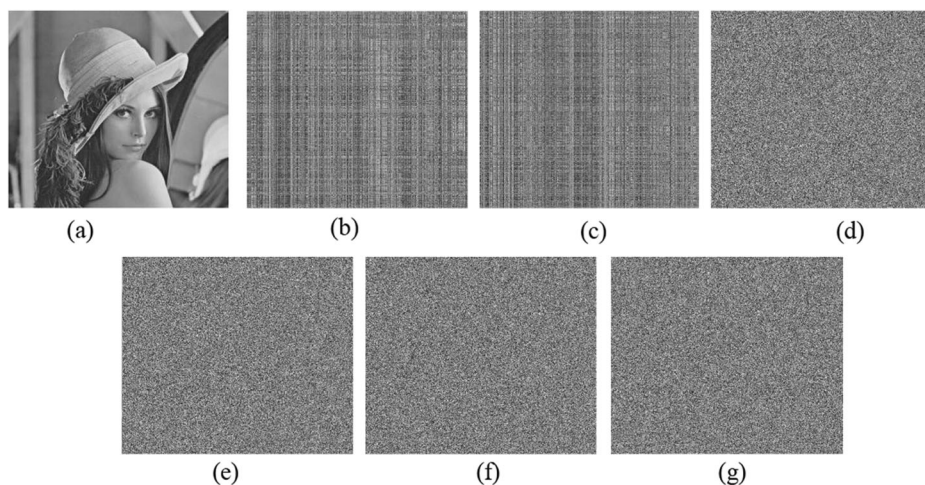


Fig. 14 Key sensitivity analysis for gray Lena image **a** Decrypted image with correct keys, **b** Decrypted image with wrong key ($b + \Delta$) of Chen system, **c** Decrypted image with wrong key ($y_0 + \Delta$) of Chen system, **d** Decrypted image with wrong key ($a + \Delta$) of Lorenz hyper system, **e** Decrypted image with wrong key ($e + \Delta$) of Lorenz hyper system, **f** Decrypted image with wrong key ($x_0 + \Delta$) of Lorenz hyper system, **g** Decrypted image with wrong key ($z_0 + \Delta$) of Lorenz hyper system

$$RMSE = \sqrt{\frac{\sum_{i=1}^M \sum_{j=1}^N [X(i,j) - Y(i,j)]^2}{M \times N}} \quad (23)$$

For better encryption/decryption performance, the values of SNR, PSNR and SSIM metrics should be as low as possible and the values of the MSE, MAE and RMSE metrics should be large. The numerical results of these metrics are shown in the Table 5. According to this table, the values of SNR, PSNR and SSIM criteria are low, which implies that the plain and encrypted images are totally different. Furthermore, the values of MSE, MAE and RMSE metrics are significantly large. Hence, the outcomes in the Table 5 confirm the security and reliability of the proposed image cryptosystem.

4.11 Key space analysis

For a good encoding method, the big key space is required in order to prevent exhaustive attack [37]. Confidential keys of the presented method are comprised of initial and system parameters values: x_0, y_0, z_0, a, b and c for Chen system, and initial as well as control parameters values: $x_0, y_0, z_0, w_0, a, b, c, d$ and e for Lorenz hyper system. It means that the overall number of secret keys is 15. If 10^{-15} is set as a computational precision, then the size of key space for the proposed image security method is $(10^{15})^{15} = 10^{225} \approx 2^{747}$ bits. This key space is sufficiently big to prohibit speculating the secret keys by brute force attack.

4.12 Key sensibility analysis

Secure image cryptosystem demands high sensibility to the confidential keys. This denotes that any tiny difference between the encryption key at the sender and its corresponding key at the receptor yields in a totally different encrypted image [10, 59]. In order to examine the sensibility of presented approach to the change in the secret keys, gray Lena image in this analysis is decrypted after modifying the initial and control parameter keys of Chen and hyper Lorenz systems by adding a very small alteration ($\Delta = 10^{-15}$). Deciphered images with correct and wrong keys are displayed in Fig. 14. Figure 14a exhibits the reconstructed image via the right keys, whereas Fig. 14b–g show the retrieved images via the incorrect keys. From this figure, it is obvious that the

Table 6 Results of differential analysis between plain and encrypted images

Images	UACI (%)	NPCR (%)
Gray Lena	33.7053	99.61
Gray Baboon	33.7850	99.61
Gray Peppers	33.7073	99.62
Boat	33.3989	99.61
Color Lena	33.0305	99.63
Color Baboon	33.9652	99.60
Color Peppers	33.7999	99.60
Barbara	33.7882	99.62
Braine	38.7230	99.57
Fingerprint	38.8963	99.62
City	33.9747	99.63

decrypted image by using a slightly modified key is extremely different from the input plain image. Hence, the restoration of the plain image with incorrect keys is quite impossible. The results in Fig. 14 reveal that the presented scheme have high sensibility to the confidential keys.

4.13 Differential analysis

In order to examine if a tenuous alteration in the input image can cause a totally different output image, two metrics are utilized in this work namely Number of Pixel Changing Rate (NPCR) and Unified Averaged Changed Intensity (UACI). These two criterions are employed to measure the plaintext sensitivity as shown below:

$$\begin{aligned} \text{NPCR} &= \frac{\sum_{i,j} D(i,j)}{M \times N} \times 100\% \\ \text{UACI} &= \frac{1}{M \times N} \left[\sum_{i,j} \frac{|C_1(i,j) - C_2(i,j)|}{255} \right] \times 100\% \\ D(i) &= \begin{cases} 0 & \text{if } C_1(i,j) = C_2(i,j) \\ 1 & \text{if } C_1(i,j) \neq C_2(i,j) \end{cases} \end{aligned} \quad (24)$$

Where C_1 and C_2 refer to the two output images whose their corresponding input images are different just by one bit, M, N point to image size and D is the difference matrix. The typical values of NPCR and UACI are about 99.6094% and 33.4635%, respectively [23, 25, 33, 40, 47]. The scores of NPCR and UACI for the sample images used in this article are listed in Table 6. It can be noticed from the computed

Table 7 Results of NIST test on chaotic sequences produced by Chen and hyper Lorenz systems

Test name	Sequence of Chen system			Sequence of hyper Lorenz system		
	X			Y		
	P value	Proportion	Result	P value	Proportion	Result
Frequency	0.5480	100/100	Success	0.6810	100/100	Success
Block frequency ($m=128$)	0.9035	100/100	Success	0.5768	100/100	Success
Runs	0.9977	100/100	Success	0.7740	100/100	Success
Longest run ($M=1000, N=100$)	0.8470	100/100	Success	0.8473	100/100	Success
Rank	0.4487	100/100	Success	0.5896	100/100	Success
FFT	0.7223	100/100	Success	0.9957	100/100	Success
Non-overlapping template ($m=9, B=001000110$)	0.2292	100/100	Success	0.4688	100/100	Success
Overlapping template ($m=9, B=110000100$)	0.8065	100/100	Success	0.6576	100/100	Success
Universal ($L=5, Q=1000, K=145000$)	0.4661	100/100	Success	0.4814	100/100	Success
Linear complexity ($M=1000$)	0.6938	100/100	Success	0.1635	100/100	Success
Serial ($M=16$)	0.9576	100/100	Success	0.6039	100/100	Success
Approximate entropy ($m=10$)	0.6404	100/100	Success	0.1968	100/100	Success
Cumulative sums (forward)	0.9997	100/100	Success	0.0321	100/100	Success
Random excursions ($x=-4$)	0.5882	100/100	Success	0.2452	100/100	Success
Random excursions variant ($x=-9$)	0.7119	100/100	Success	0.6434	100/100	Success

outcomes of Table 6 that the NPCR and UACI values are extremely near to their ideal values. This indicates that the presented approach is so sensitive to the plaintext image and thereby it can attain high level performance in resisting differential analysis.

4.14 NIST test analysis

This test is utilized to measure the randomness of the produced binary sequences. Statistical NIST test comprises of fifteen different tests. The basic idea for all these tests is based on a comparison between α and P , where α refers to the decision rule or significance level and α is set to 0.01, whilst P points to the probabilistic value. If the calculated P – value for the binary sequence at each test is larger than α , then the generated sequence passes the NIST tests and it is random with a 99% of confidence. Else, the sequence cannot pass the NIST tests and it is not random with a 99% of confidence [53]. 100 binary series with a stream length of 100,000 bits are employed to perform the 15 NIST tests in this article and the outcomes are shown in the Table 7. It can be found from Table 7 that the obtained sequences from Chen and hyper Lorenz systems used in the presented scheme possess high randomness and can pass successfully each test of statistical NIST suite.

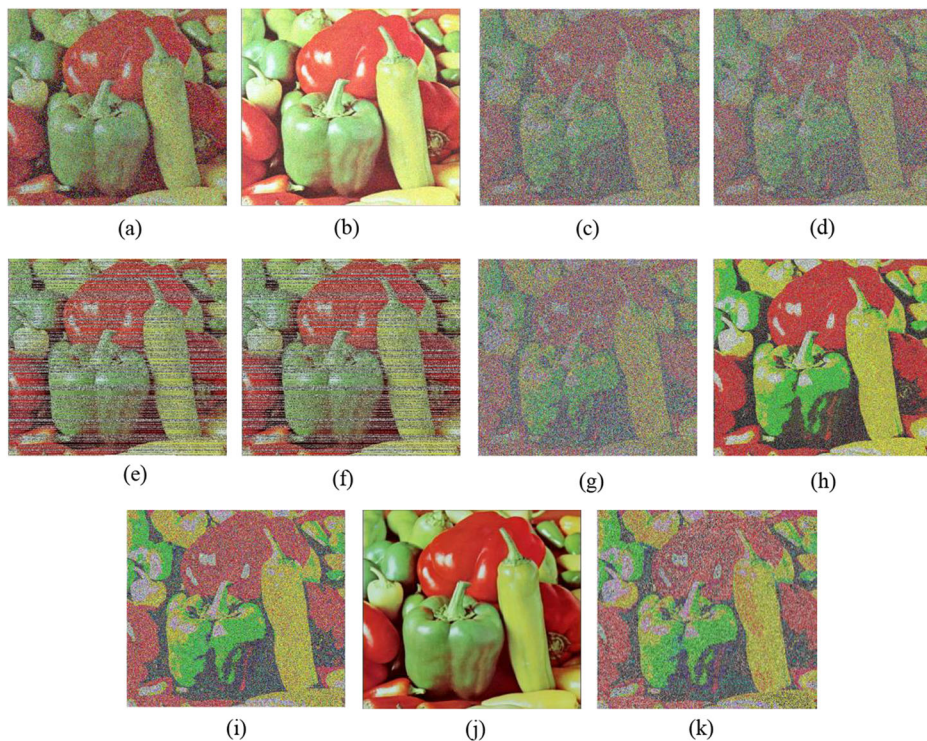


Fig. 15 The results of image processing attacks analysis on the decrypted image **a** Adding salt & pepper noise with density 0.4, **b** Adding Gaussian noise with density 0.2, **c** Adding speckle noise with density 0.01, **d** Adding Poisson noise, **e** Cropping (50%), **f** Jpeg Compression (QF = 10), **g** Low pass filter (3*3), **h** Sharpening, **i** Blurring ($len = 10$, $\vartheta = 45$), **j** Histogram equalization, **k** Gamma correction ($\gamma = 0.5$)

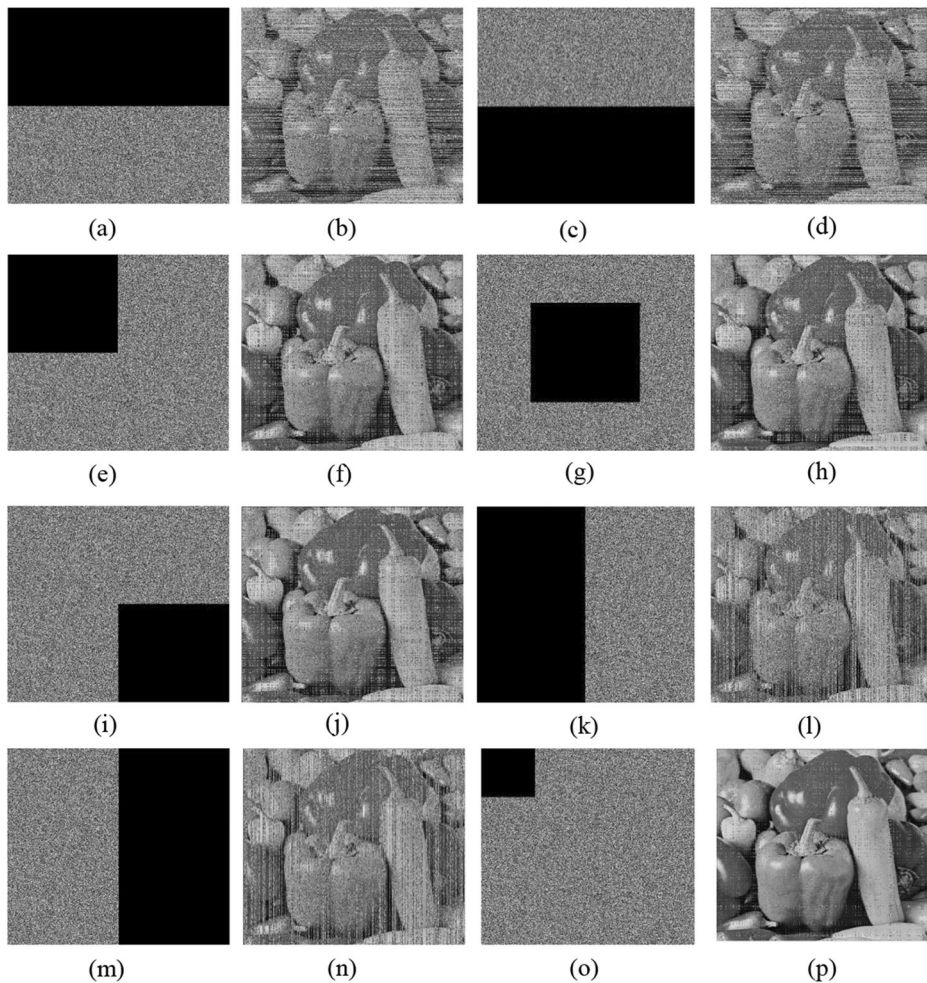


Fig. 16 The results of cropping attack on the encrypted and decrypted images **a** Cropping 50% from top of the encrypted image, **b** Decrypted image of (a), **c** Cropping 50% from bottom of the encrypted image, **d** Decrypted image of (c), **e** Cropping 25% from top left of the encrypted image, **f** Decrypted image of (e), **g** Cropping 25% from center of the encrypted image, **h** Decrypted image of (g), **i** Cropping 25% from bottom right of the encrypted image, **j** Decrypted image of (i), **k** Cropping 50% from left of the encrypted image, **l** Decrypted image of (k), **m** Cropping 50% from right of the encrypted image, **n** Decrypted image of (m), **o** Cropping 12.5% from top left of the encrypted image, **p** Decrypted image of (o)

4.15 Immunity to image processing attacks analysis

When the encrypted image is transferred throughout the telecommunication channels, it can be demolished by the intruders. This means that the encrypted image cannot reach correctly to the intended receptor. Therefore, the important parameter for a typical cryptosystem is immunity to cryptographic attacks [8, 68, 70]. To clarify the immunity of presented approach against some known image processing attacks, the enciphered image in this test is attacked by different sorts of noises with different intensities including salt & pepper, Gaussian, speckle and Poisson noises, cropping pixels from different image sides, Jpeg compression, low pass filter,

Table 8 The PSNR, SNR and SSIM scores for the decrypted image after adding image processing attacks

Attacks	Color Lena		Gray Baboon		Color Peppers		Fingerprint		City	
Salt & pepper ($\sigma^2=0.4$)	PSNR	12.5947	PSNR	13.5288	PSNR	12.0519	PSNR	10.4022	PSNR	13.0892
	SNR	7.4572	SNR	8.0760	SNR	5.8774	SNR	8.4040	SNR	6.6128
	SSIM	0.2504	SSIM	0.3914	SSIM	0.2348	SSIM	0.6099	SSIM	0.4390
Salt & pepper ($\sigma^2=0.2$)	PSNR	15.5965	PSNR	16.5085	PSNR	15.0512	PSNR	13.4138	PSNR	16.0886
	SNR	10.4591	SNR	11.0708	SNR	8.8768	SNR	11.4156	SNR	9.6121
	SSIM	0.3975	SSIM	0.5961	SSIM	0.3757	SSIM	0.7540	SSIM	0.6411
Salt & pepper ($\sigma^2=0.1$)	PSNR	18.6087	PSNR	19.5319	PSNR	18.0387	PSNR	16.3877	PSNR	19.1337
	SNR	13.4713	SNR	14.0943	SNR	11.8642	SNR	14.3895	SNR	12.6573
	SSIM	0.5425	SSIM	0.7432	SSIM	0.5160	SSIM	0.8213	SSIM	0.7833
Gaussian noise ($\sigma^2=0.2$)	PSNR	13.6529	PSNR	13.2005	PSNR	13.3229	PSNR	16.4802	PSNR	13.2445
	SNR	8.5154	SNR	7.7628	SNR	7.1484	SNR	14.4820	SNR	6.7681
	SSIM	0.5835	SSIM	0.7254	SSIM	0.5214	SSIM	0.8972	SSIM	0.7489
Gaussian noise ($\sigma^2=0.1$)	PSNR	17.4015	PSNR	17.0472	PSNR	17.1411	PSNR	19.9683	PSNR	17.1309
	SNR	12.2641	SNR	11.6096	SNR	10.9666	SNR	17.9701	SNR	10.6545
	SSIM	0.5888	SSIM	0.7534	SSIM	0.5494	SSIM	0.9186	SSIM	0.7945
Gaussian noise ($\sigma^2=0.01$)	PSNR	20.1731	PSNR	19.9594	PSNR	20.2470	PSNR	21.4564	PSNR	20.0418
	SNR	15.0357	SNR	14.5218	SNR	14.0725	SNR	19.4582	SNR	13.5653
	SSIM	0.5919	SSIM	0.7632	SSIM	0.5821	SSIM	0.9388	SSIM	0.8147
Speckle noise ($\sigma^2=0.01$)	PSNR	4.7721	PSNR	10.3953	PSNR	4.2602	PSNR	2.5966	PSNR	5.2621
	SNR	4.4059	SNR	4.9576	SNR	2.8570	SNR	5.3695	SNR	3.5569
	SSIM	0.0958	SSIM	0.1663	SSIM	0.0845	SSIM	0.2192	SSIM	0.1667
Poisson noise	PSNR	9.5742	PSNR	10.4111	PSNR	9.0647	PSNR	7.4037	PSNR	10.0567
	SNR	4.4368	SNR	4.9734	SNR	2.8902	SNR	5.4054	SNR	3.5803
	SSIM	0.0992	SSIM	0.1665	SSIM	0.0866	SSIM	0.2254	SSIM	0.1707
Cropping (12.5%)	PSNR	20.6728	PSNR	21.6759	PSNR	20.1083	PSNR	18.4816	PSNR	21.1599
	SNR	15.5354	SNR	16.2382	SNR	13.9338	SNR	16.4834	SNR	14.6835
	SSIM	0.6666	SSIM	0.8315	SSIM	0.6458	SSIM	0.8560	SSIM	0.8563
Cropping (25%)	PSNR	14.6317	PSNR	15.5520	PSNR	14.1198	PSNR	12.5218	PSNR	15.1516
	SNR	9.4942	SNR	10.1143	SNR	7.9453	SNR	10.5236	SNR	8.6752
	SSIM	0.3474	SSIM	0.5270	SSIM	0.3264	SSIM	0.6952	SSIM	0.5733
Cropping (50%)	PSNR	11.6188	PSNR	12.5324	PSNR	11.0985	PSNR	9.4698	PSNR	12.1272
	SNR	6.4813	SNR	7.0948	SNR	4.9240	SNR	7.4715	SNR	5.6507
	SSIM	0.1923	SSIM	0.3063	SSIM	0.1803	SSIM	0.4875	SSIM	0.3390
Jpeg Compression (QF=10)	PSNR	9.4430	PSNR	9.5310	PSNR	8.9268	PSNR	6.5608	PSNR	12.1272
	SNR	4.3055	SNR	4.0933	SNR	2.7523	SNR	4.5626	SNR	5.6507
	SSIM	0.0448	SSIM	0.0334	SSIM	0.0359	SSIM	0.0126	SSIM	0.3390
Low pass filter (3*3)	PSNR	9.3725	PSNR	10.2793	PSNR	8.8302	PSNR	7.0959	PSNR	9.8916
	SNR	4.2350	SNR	4.8416	SNR	2.6557	SNR	5.0977	SNR	3.4152
	SSIM	0.0890	SSIM	0.1698	SSIM	0.0750	SSIM	0.1696	SSIM	0.1596
Sharpening	PSNR	13.2732	PSNR	13.5763	PSNR	12.9994	PSNR	11.5491	PSNR	13.5293
	SNR	8.1358	SNR	8.1387	SNR	6.8249	SNR	9.5508	SNR	7.0529
	SSIM	0.2660	SSIM	0.4774	SSIM	0.2402	SSIM	0.6434	SSIM	0.4711
Blurring ($len=10, \vartheta=45$)	PSNR	8.7400	PSNR	9.6346	PSNR	8.1958	PSNR	6.5218	PSNR	9.2391
	SNR	3.6026	SNR	4.1970	SNR	2.0213	SNR	4.5236	SNR	2.7627
	SSIM	0.0463	SSIM	0.0575	SSIM	0.0383	SSIM	0.0366	SSIM	0.0536
Histogram equalization	PSNR	44.1461	PSNR	44.1444	PSNR	44.0888	PSNR	44.2556	PSNR	44.1408
	SNR	39.0087	SNR	38.7067	SNR	37.9143	SNR	42.2573	SNR	37.6643
	SSIM	0.9957	SSIM	0.9984	SSIM	0.9941	SSIM	0.9990	SSIM	0.9989
Gamma correction ($\gamma=0.5$)	PSNR	10.6568	PSNR	12.0197	PSNR	9.9161	PSNR	8.2963	PSNR	11.2545
	SNR	5.5193	SNR	6.5821	SNR	3.7417	SNR	6.2981	SNR	4.7781
	SSIM	0.1664	SSIM	0.3416	SSIM	0.1328	SSIM	0.3637	SSIM	0.2978

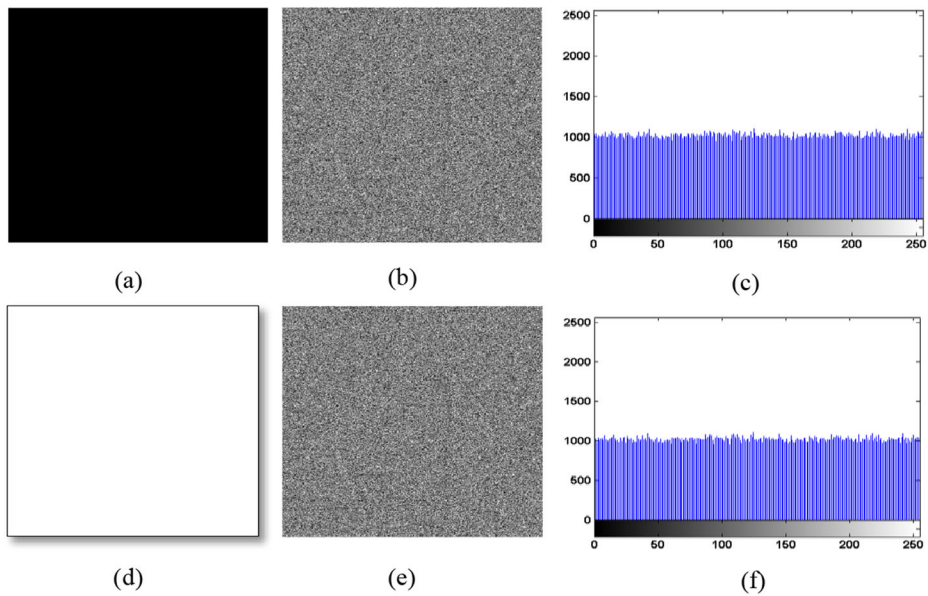


Fig. 17 Results of known plaintext/chosen plaintext analysis **a** Black image, **b** Encrypted image of (a), **c** Histogram of (b), **d** White image, **e** Encrypted image of (d), **f** Histogram of (e)

sharpening, blurring, histogram equalization and gamma correction. Furthermore, in order to demonstrate the robustness of encrypted image against these attacks, the PSNR, SNR and SSIM visual metrics are used to quantify the quality decrypted image at the recipient. The computed outcomes of PSNR, SNR and SSIM for the five test images after adding the earlier mentioned attacks with various parameters are given in Table 8. The reconstructed color Peppers image after applying the attacks on the encrypted image is presented in Fig. 15. Moreover, the effect of cropping attack is assessed for encrypted and decrypted images after cutting regions with different sizes and different locations from gray Peppers image, and the results are displayed in Fig. 16. It is revealed from Table 8, Figs. 15 and 16 that despite the strong attacks on the encrypted image, the quality of the decrypted image is still acceptable, which proves the immunity of the proposed scheme against some popular image processing attacks.

4.16 Known plaintext/chosen plaintext attack analysis

Known plaintext and chosen plaintext analyses are among the strongest common attacks in the field of cryptography. Usually, the opponent tries a special sort of image like black or white images in an attempt to attack the cryptographic schemes to obtaining the secret keys [14]. To defeat this attack, our proposed approach encrypts black and white images as shown in the Fig. 17. It is clear from this figure that the encrypted images in Fig. 17b and e, respectively contain no meaningful information and their corresponding histograms in Fig. 17c and f, respectively are quite uniform. This proves that the presented method can modify and encrypt all the pixels in black and white images and hence it can counter these two attacks efficiently.

Table 9 Encryption time of the test images

Images (512 × 512)	Encryption time (s)
Gray Lena	1.634973
Gray Baboon	1.646432
Gary Peppers	1.645470
Boat	1.649609
Images (512 × 512)	Encryption time (s)
Color Lena	2.017933
Color Baboon	2.329260
Color Peppers	2.180964
Barbara	2.129236
Images (256 × 256)	Encryption time (s)
Braine	0.548691
Fingerprint	0.545923
City	0.531774

4.17 Computational speed analysis

Time cost represents a significant factor to measure the cryptosystem efficiency in real time communication applications [11, 14, 23]. The software and hardware environments to implement the numerical and visual experiments of the proposed algorithm were mentioned in Section 4. Initially, the encryption process starts by dividing the input image into two images. In the second step, the key stream of Chen system is produced to realize the permutation operation; this step is followed by key stream generation of hyper Lorenz system to carry out the diffusion operation in the third step. Now, after completing permutation/diffusion procedure to get the two encrypted images, image encoding using DNA computing rules is take apart in the fourth step before employing DNA addition process in the fifth step to gain the encoded DNA image. The eventual step is accomplished by applying DNA decoding rules to obtain the ultimate ciphered image. The total consumed time to execute these six steps for the test grayscale images (Lena, Baboon, Peppers and Boat), of size (512 × 512), and color images (Lena, Baboon, Peppers and Barbara) of size (512 × 512), as well as the three other images (Braine, Fingerprint and City) of size (256 × 256) is illustrated in Table 9. According to this table, the time consumed to encrypt grayscale images is lower than that for color images. For example, the total encryption times for gray Lena, gray Baboon and gray Peppers images of size (512 × 512) are 1.634973, 1.646432 and 1.645470 s, respectively. On the other hand, the total encryption time for color Lena, color Baboon and color Peppers images with the same size are 2.017933, 2.329260 and 2.180964 s, respectively. In addition, the consumed time for encryption increases as the input image size increases. For instance, the encryption time for Braine image of size (256 × 256) is 0.548691 s, whilst the encryption time for Boat and Barbara images of size (512 × 512) are 1.649609 and 2.129236 s, respectively. In addition, based on the results of Table 9, the time consumption of encryption operation is short which indicates that the presented technique possesses excellent processing speed for encrypting different sorts of images with various sizes. Thus, the suggested approach is a suitable candidate for practical applications of image encryption over open communication networks.

Table 10 Entropy, correlation in three directions, NPCR and UACI comparison results

Image	Scheme	Global entropy	r_{xy}			NPCR (%)	UACI (%)
			H	V	D		
Gray Lena	Ref. [13]	7.9993	−0.0045	−0.000162	0.0053	99.62	33.45
	Ref. [32]	7.9993100	0.0013	0.0008	0.0066	99.61	33.48
	Ref. [57]	7.9994	0.0032	0.0016	0.0023	99.6002	33.5079
	Ref. [53]	7.999303	0.0027	0.0003	0.0012	99.6009	33.4330
	Ref. [48]	7.9992	−0.0685	0.0857	0.0059	99.60	33.39
	Ref. [60]	7.9993	0.00047	−0.03911	0.00305	99.6641	33.6124
	Ref. [45]	7.99642	0.00051	−0.0043	0.00031	99.47	33.23
	Ref. [12]	7.9993	0.0044	0.0151	0.0012	99.62	33.45
	Ref. [10]	7.9993	−0.0285	0.0014	0.0013	99.62	33.46
	Ref. [33]	7.9993	0.0023	0.0009	0.0008	99.6103	33.4504
	Ref. [14]	7.9993	−0.0139	0.0177	0.00067947	99.58	33.43
	Ref. [34]	7.9993	−0.0013	0.0005	0.0016	99.6161	33.4709
	Proposed	7.9993	−0.0215	−0.0113	0.0089	99.61	33.7053
Gray Baboon	Ref. [13]	7.9993	−0.0108	0.0087	0.0058	99.61	33.43
	Ref. [11]	7.9993	—	—	—	99.61	33.43
	Ref. [32]	7.9992644	0.0050	0.0015	0.0016	99.6210	33.4308
	Ref. [57]	7.9992	0.0029	0.0033	0.0062	99.5903	33.5281
	Ref. [53]	—	0.00005	0.0007	0.0051	99.601	33.4538
	Ref. [60]	7.9993	0.00318	0.00285	−0.00294	99.6438	33.6430
	Ref. [69]	7.99923	−0.0022	−0.0064	−0.0023	99.596	33.459
	Ref. [45]	7.99645	−0.00245	−0.00314	0.00137	99.36	33.21
	Ref. [12]	7.9993	−0.0058	0.0131	0.0030	99.63	33.41
	Ref. [10]	7.9993	0.0013	−0.0281	0.0128	99.61	33.42
	Ref. [33]	7.9994	0.0019	0.0005	−0.0023	99.6047	33.4704
	Ref. [14]	7.9994	−0.0106	0.0036	0.0180	99.63	33.41
	Ref. [34]	7.9993005	0.0032	0.0012	−0.0004	99.6109	33.4792
Gray Peppers	Proposed	7.9993	0.0141	−0.0075	0.0027	99.61	33.7850
	Ref. [11]	7.9994	—	—	—	99.61	33.52
	Ref. [32]	7.9993762	−0.0053	0.0056	0.0018	99.6028	33.5033
	Ref. [57]	7.9993	0.0006	0.0038	0.0010	99.6112	33.5265
	Ref. [53]	—	0.0019	0.0028	0.0019	99.6437	33.4794
	Ref. [60]	7.9994	0.00198	0.04321	0.02547	99.6287	33.6012
	Ref. [69]	7.99937	−0.0020	−0.0046	−0.0070	—	—
	Ref. [45]	7.98953	−0.00178	0.00812	−0.00031	99.52	33.34
	Ref. [10]	7.9993	−0.0282	0.0100	−0.0012	99.62	33.44
	Ref. [33]	7.9993	0.0020	−0.0010	0.0011	99.6099	33.4618
	Proposed	7.9993	0.0041	0.000057955	0.00097232	99.62	33.7073
	Ref. [23]	7.999273	—	—	—	99.601364	33.456980
	Ref. [32]	7.9993384	−0.0078	−0.0012	−0.0032	99.6068	33.4881
Boat	Ref. [57]	7.9994	0.0003	0.0034	0.0011	99.6124	33.5891
	Ref. [69]	7.99937	0.0060	0.0021	−0.0061	99.611	33.514
	Proposed	7.9993	−0.0031	0.0166	−0.0086	99.61	33.3989
Color Lena	Ref. [44]	7.9982	0.000187	0.000592	−0.000736	99.61	33.45
	Ref. [43]	—	—	—	—	99.6249	33.2404
	Ref. [19]	7.999772	0.01376	0.0024	−0.00353	99.619	33.447
	Ref. [51]	7.9993	−0.0034	0.00193	−0.0134	99.62	33.476
	Ref. [50]	7.9992	0.0004	0.0003	−0.0005	99.82	33.3354
	Ref. [27]	7.9994	−0.0043	−0.0007	0.0030	99.6037	33.4488
	Ref. [8]	7.9993	−0.0031	−0.0293	0.0077	99.6100	33.4515
	Ref. [67]	7.997	0.0076	−0.0125	0.0101	—	—
	Proposed	7.9994	0.00071219	−0.0027	−0.0104	99.63	33.0305
Color Baboon	Ref. [43]	—	—	—	—	99.7091	33.3056
	Ref. [19]	7.999778	−0.0192	−0.00503	0.0138	99.609	33.457
	Ref. [50]	7.9991	−0.0009	−0.0007	0.0006	99.7885	33.3452

Table 10 (continued)

Image	Scheme	Global entropy	r_{xy}			NPCR (%)	UACI (%)
			H	V	D		
Color Peppers	Ref. [27]	7.9993	−0.0084	−0.0015	−0.0024	99.6147	33.4470
	Ref. [8]	7.9993	0.0224	0.0115	−0.0025	99.6111	33.4812
	Ref. [67]	7.9826	−0.0042	0.0099	0.0050	99.5937	33.3803
	Proposed	7.9993	−0.0143	−0.0174	0.0091	99.60	33.9652
	Ref. [43]	—	—	—	—	99.7065	33.3247
	Ref. [19]	7.999788	0.00123	0.00596	−0.0046	99.608	33.466
	Ref. [51]	7.9994	−0.00083	−0.000066	0.00293	—	—
	Ref. [50]	7.9989	−0.0011	0.0013	0.0015	99.8446	33.3321
	Ref. [27]	7.9994	0.0006	0.0065	−0.0011	99.6040	33.4600
	Ref. [8]	7.9993	0.0282	0.0122	−0.0056	99.6240	33.4487
Barbara	Ref. [67]	7.9908	0.0080	0.0062	0.0070	99.6126	33.4151
	Proposed	7.9993	0.00027303	−0.0055	0.0142	99.60	33.7999
	Ref. [50]	7.9992	−0.0036	−0.0029	0.0018	99.7369	33.3417
	Ref. [27]	7.9994	−0.0014	0.0075	0.0116	99.6021	33.4762
	Proposed	7.9992	−0.0058	−0.0117	0.0137	99.62	33.7882

4.18 Computational complexity analysis

Computational complexity analysis is one of the important tests to evaluate the efficiency of encryption algorithms. The computation cost is related to the encryption stages [26, 56]. First, the algorithm applies bit replacing technique on the input image, the proposed method needs $O(M \times N)$ iterations in this stage, where M, N represent the size of the input image. Secondly,

Table 11 Discrepancy metrics comparison results

Image	Scheme	SNR (dB)	MSE	PSNR (dB)	SSIM	MAE	RMSE
Gray Lena	Ref. [60]	—	7747.309	9.23929	—	—	—
	Proposed	3.5545	7797.7	9.2111	0.0350	73.1986	88.3047
Gray Baboon	Ref. [60]	—	7254.201	9.52486	—	—	—
	Proposed	4.0993	7234.2	9.5369	0.0339	70.8518	85.0539
Gray Peppers	Ref. [60]	—	8413.235	8.88117	—	—	—
	Proposed	2.8991	8472.7	8.8506	0.0333	75.7536	92.0474
Color Lena	Ref. [44]	—	39.101	8.6728	0.0341	—	—
	Ref. [50]	—	—	8.583804	0.021284	—	—
	Ref. [27]	2.2281	8940.46	8.6170	0.0100	77.5708	94.5564
	Ref. [8]	—	—	8.6157	0.0100	—	—
	Ref. [26]	—	10,116.2	8.0935	—	—	—
	Ref. [30]	3.9755	5812.68	10.2354	—	64.2491	78.8175
Color Baboon	Proposed	5.1582	10,637	7.8625	0.0331	84.2279	103.1376
	Ref. [50]	—	—	8.781472	0.019615	—	—
	Ref. [27]	2.2193	8628.31	8.7715	0.0090	76.3645	92.8887
	Ref. [8]	—	—	8.7814	0.0091	—	—
Color Peppers	Proposed	4.0527	8635.6	8.7679	0.0358	76.4112	92.9280
	Ref. [50]	—	—	8.079685	0.019063	—	—
	Ref. [27]	1.6411	10,128.01	8.0756	0.0091	82.2445	100.6380
	Ref. [8]	—	—	8.0812	0.0093	—	—
Barbara	Proposed	4.6372	7869.4	9.1714	0.0415	73.4397	88.7094
	Ref. [50]	—	—	8.740601	0.018851	—	—
	Ref. [27]	1.7446	8526.60	8.8230	0.0089	75.9906	92.3396
	Proposed	3.7307	8516.2	8.8284	0.0306	75.9600	92.2830

Table 12 Key space comparison results

Scheme	Precision	Secret key space
Ref. [23]	10^{-14}	10^{56}
Ref. [44]	—	2^{189}
Ref. [17]	10^{-16}	2^{213}
Ref. [13]	10^{-14}	5.46×10^{80}
Ref. [55]	10^{-14}	2^{233}
Ref. [56]	10^{-15}	2^{299}
Ref. [11]	10^{-14}	10^{70}
Ref. [41]	10^{-16}	2^{138}
Ref. [32]	10^{-14}	2^{428}
Ref. [57]	10^{-14}	10^{112}
Ref. [65]	10^{-16}	10^{160}
Ref. [50]	10^{-14}	10^{168}
Ref. [46]	10^{-15}	2^{662}
Ref. [53]	10^{-14}	2^{440}
Ref. [8]	10^{-14}	2^{232}
Ref. [58]	10^{-14}	2^{128}
Ref. [26]	—	2^{299}
Ref. [69]	10^{-14}	10^{70}
Ref. [12]	—	10^{90}
Ref. [10]	10^{-10}	2^{256}
Ref. [59]	10^{-16}	2^{300}
Ref. [33]	10^{-15}	2^{249}
Ref. [70]	10^{-15}	2^{296}
Ref. [68]	10^{-14}	10^{56}
Ref. [14]	10^{-14}	10^{98}
Ref. [34]	10^{-15}	2^{455}
Ref. [67]	—	6×2^{192}
Proposed	10^{-15}	$10^{225} \approx 2^{747}$

Table 13 SSIM and PSNR comparison results under image processing attacks

Attacks	Color Lena		Color Peppers	
	SSIM		SSIM	
	Ref. [50]	Proposed	Ref. [50]	Proposed
Gaussian noise ($\sigma^2=0.01$)	0.20007	0.5919	0.18439	0.5821
Sharpening	0.15691	0.2660	0.14338	0.2402
Blurring	0.04248	0.0463	0.04006	0.0383
Attacks	PSNR (dB)		PSNR (dB)	
	Ref. [8]	Proposed	Ref. [8]	Proposed
Salt & pepper ($\sigma^2=0.1$)	18.5825	18.6087	18.0163	18.0387
Gaussian noise ($\sigma^2=0.01$)	14.6608	20.1731	14.1389	20.2470
Cropping (25%)	14.5775	14.6317	14.0297	14.1198
Jpeg Compression (QF=10)	9.3480	9.4430	8.7830	8.9268
Sharpening	13.2477	13.2732	13.0321	12.9994
Blurring	9.4418	8.7400	8.8964	8.1958
Histogram equalization	30.7128	44.1461	32.1093	44.0888
Gamma correction ($\gamma=0.5$)	10.6780	10.6568	9.8371	9.9161
Attacks	Color Lena		Color Peppers	
	SSIM		PSNR (dB)	
	Ref. [27]	Proposed	Ref. [27]	Proposed
Salt & pepper ($\sigma^2=0.1$)	0.3287	0.5425	25.54	18.6087
Salt & pepper ($\sigma^2=0.2$)	0.2651	0.3975	17.59	15.5965
Gaussian noise ($\sigma^2=0.1$)	0.0851	0.5888	12.99	17.4015
Gaussian noise ($\sigma^2=0.2$)	0.0661	0.5835	11.25	13.6529

Table 14 Comparison results of encryption times

Image (512 × 512)	Scheme	Encryption time (s)
Gray Lena	Ref. [23]	0.226201
	Ref. [53]	0.06910
	Ref. [48]	2.792
	Ref. [45]	19.2
	Ref. [12]	1.26
	Ref. [64]	0.0270
	Ref. [10]	13.02
	Proposed	1.634973
Gray Baboon	Ref. [45]	18.7
	Proposed	1.646432
Gray Peppers	Ref. [54]	1.067064
	Ref. [45]	19.5
	Proposed	1.645470
Color Lena	Ref. [19]	0.062
	Ref. [50]	3.768142
	Ref. [49]	8.41
	Proposed	2.017933
Color Baboon	Ref. [50]	3.780993
	Proposed	2.329260
Color Peppers	Ref. [50]	3.741908
	Proposed	2.180964
Barbara	Ref. [50]	3.766197
	Proposed	2.129236

the generation of chaotic sequences from Chen and Lorenz's hyper systems needs also $O(M \times N)$ iterations. Next, the permutation and diffusion processes require $O(2 \times M \times N)$ iterations. After obtaining the diffused image, the computation complexity of employing DNA coding/addition operations upon the resultant image is $O(2 \times M \times N)$. Then, the decoding DNA process of the encoded image to get the final encrypted image needs $O(M \times N)$. Thus, the total computational complexity of the presented scheme is $O(7 \times M \times N)$. The computational complexity of the proposed cryptosystem is lower than Ref. [56]. In addition, the time complexity of the proposed method is lower as compared with most algorithms according to the computational time results in Tables 9 and 14. It can be revealed from the above analysis that the proposed approach is more efficient in comparison with some similar current image encryption mechanisms.

5 Performance comparative analysis

The comparative analysis results are presented in this section by comparing our proposed approach with existing approaches in the literature in terms of security and efficiency. Tables 10, 11, 12, 13 and 14 are used to show the results in terms of entropy, correlation, NPCR, UACI, discrepancy metrics, key space, quality measures under different attacks and the total encryption times. Table 10 is presented to show the results for the grayscale Lena image, the entropy of the proposed method is approximately equal to those in other methods except for Ref. [50], also the correlation coefficients in three directions are lower than the compared schemes, and the obtained UACI and NPCR values are higher except for Refs. [10, 12, 13, 60]. For grayscale Baboon image, our approach achieves larger entropy except for Ref.

[14, 33], the correlation scores produced via the presented technique are smaller, while UACI and NPCR outcomes are higher than those in other techniques except for Refs. [12, 14, 32, 60]. For grayscale Peppers image,

The described method gives better results in terms of entropy and correlation coefficients except for Refs. [11, 60], and the gained NPCR and UACI results are also better except for Ref. [53]. For Boat image, the obtained entropy, correlation, NPCR and UACI scores are larger than those in other approaches except for Refs. [23, 32, 57, 69]. For color Lena image, all the comparative measures are higher than the existing methods except for UACI metric. For color Baboon image, the suggested cryptosystem has higher scores for the quality metrics except for NPCR outcome in comparison with the previous algorithms. For color Peppers image, the obtained entropy, correlation, UACI and NPCR values are larger except for the corresponding entropies and NPCR values in Refs. [8, 27, 43, 50, 51, 67]. Finally, for Barbara image, the presented method achieves higher performance than the compared methods except for the NPCR value in Ref. [50] and the entropy value in Ref. [27]. Furthermore, for grayscale Lena, Baboon and Peppers images in Table 11, the obtained MSE are higher, whereas PSNR outcomes are smaller than the current mechanisms. Moreover, SNR results achieved by the present algorithms in the same table are lower than values obtained via this mechanism for color Lena, Baboon, Peppers and Barbara images. On the other hand, MSE scores for gray and color images are bigger except for Ref. [27]. Also, PSNR scores are smaller except for Refs. [8, 27, 50] in the case of color Lena image. However, the existing approaches achieves lower SSIM outcomes. MAE values gained by this mechanism are better except for Ref. [27] and eventually, RMSE results of the presented work are larger except for Ref. [27].

According to Table 12, the suggested mechanism possesses the largest key space size in the comparison with current mechanisms. Table 13 illustrates the comparison results under different image processing attacks for color Lena and Peppers images by the means of SSIM and PSNR measures. The proposed method achieves better SSIM results than Ref. [50] in Gaussian noise addition, sharpening and blurring attacks. Additionally, PSNR outcomes produced via this work are higher than Ref. [8] in the compared attacks except for blurring and gamma correction attacks on Lena image, and sharpening and blurring attacks on Peppers image. Furthermore, this cryptosystem possesses larger SSIM and PSNR values under salt and pepper, and Gaussian noise attacks with various densities as compared with Ref. [27] for Lena image except for PSNR scores in the case of salt and pepper noise addition. This means that the recovering ability of image with good quality for this method under noise attack is better. Ultimately, Table 14 gives the comparative results of total encryption time for the test gray and color images. The Ref. [12, 23, 53, 64] are faster than our method for gray Lena image. On the other hand, the proposed method runs faster than Ref. [45] for gray Baboon image. In the case of color images, our approach is faster than Refs. [49, 50] and slower than Ref. [19]. Based on the previous comparison analysis and the results in Tables 10, 11, 12, 13 and 14, we can say that the presented image cryptosystem possesses superior performance concerning the entropy, correlation coefficient, NPCR, UACI, SNR, MSE, PSNR, SSIM, MAE, RMSE, key space and encryption time in comparison with state of art methods. Hence, the proposed scheme is an effective for image encryption technology.

6 Conclusions

A new image cryptosystem that based on bit replacing technique, chaotic systems and DNA sequence computations is proposed in this article. Bit replacing technique is one of the contributions of this research which can fulfill the desired tradeoff between security and rapidity. First of all, the original image is divided into different independent images by employing bit replacing technique. Secondly, in order to break the robust relationship among neighboring image pixels by modifying the locations and values of image pixels, chaotic sequences produced by 3D Chen system and 4D hyper Lorenz system are adopted based on the principle of permutation-diffusion. Therefore, the presented approach can conquer the common shortcomings of other methods in the literature that based on low dimension chaotic system by utilizing higher dimensional and hyper chaotic systems. Thirdly, the encoding and decoding rules of DNA technology are exploited to strengthen the proposed system security. Eventually, DNA addition operation is implemented to integrate the resultant encoded DNA images and complete the encryption process. Performance analysis is carried out in order to verify the effectiveness of the proposed method including spatial distribution analysis, histogram analysis, global and local entropy analysis, correlation analysis, discrepancy metrics analysis, key space and key sensitivity analysis, differential analysis, NIST analysis, robustness to common attacks, speed analysis and computational complexity analysis. The computed results indicate that the presented approach possesses high key sensibility, big key space, low correlation coefficient, uniform histogram, ideal entropy, ideal NPCR and UACI values, high randomness, and short encryption time. In addition, this approach can endure popular attacks like statistical, brute force, differential, known plaintext/chosen plaintext attacks as well as most common image processing attacks. Also, this cryptosystem is compared with state of art approaches and the results manifest that the suggested approach outperforms some compared methods by the means of security and efficiency performance. Thus, the proposed approach can be applied to encrypt and decrypt different sorts of images with different sizes such as grayscale image, RGB image, medical image, biometric image and remote sensing image. Hence, it can be employed to protect image information in real time telecommunication channels. As a future work, this image cryptosystem can be applied on multimedia data such as video and audio signals. Besides, more researches can be conducted to improve the computational complexity without loss of performance and security. Moreover, the presented mechanism can be also implemented on FPGA hardware system.

Declarations

Conflict of interest The authors confirm that the manuscript has been submitted solely to this journal and is not published, in press, or submitted elsewhere. This work was not supported by grants or funding. There are no financial interests or benefits dependent on this manuscript. There is also no conflict of interest with any other research.

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